University at Buffalo
The State University of New York

International CP2K-Newton-X-Libra-WaveMixings Autumn School

Alexey Akimov[1] and Mohammad Shakiba[1,2]

[1]University at Buffalo, SUNY
[2]University of California, Merced

October 15, 2026
Kiel, Germany (virtual)
5 pm – Kiel; 11 am – Buffalo; 8 am – Merced

https://lms.uni-kiel.de/url/RepositoryEntry/5837357358

University at Buffalo
The State University of New York

Part 1: Brief Libra Overview

Libra as a workhorse of our developments and applied studies

University at Buffalo
The State University of New York



- Modular, open-source methodology “discovery”/prototyping library for nonadiabatic and quantum dynamics (NAQD)
- Yet a software for practical simulation of NAQD in large atomistic systems

Akimov *JCC,* **2016**, 37, 1626

https://github.com/Quantum-Dynamics-Hub/libra-code

Shakiba, M.; Smith, B.; Li, W.; Dutra, M.; Jain, A.; Sun, X.; Garashchuk, S.; Akimov, A.V. "Libra: A modular software library for quantum nonadiabatic dynamics" *Software Impacts* **2022**, 14, 100445

Latest Release Libra v5.12.0 DOI 10.5281/zenodo.19162602

Nonadiabatic dynamics:

- Exact: finite-difference, DVR (Colbert-Miller), SOFT
- TD-SE and Liouville equation
- Initialization and TD-SE integration: in adiabatic and diabatic rep
- Surface hopping and SH/SE populations: in adiabatic and diabatic rep
- Ehrenfest, FGR, HEOM, QTAG, ETSH/ETSH-3
- TSH methods: FSSH, GFSH, MSSH, BCE, FSSH-2, FSSH-3, MASH, QTSH, AFSSH, MFSD, etc.)
- Exact Factorization (XF) methods: SHXF, XFMF, MQCXF
- Decoherence (mSDM, EDC, ID-A, ID-F, DISH, DISH_rev2023, BCSH, Shenvi-Subotnik-Yang correction, Zhu correction for QCLE)
- Decoherence times: user-defined, Schwartz, Subotnik, XF, dephasing-informed correction, etc.
- Integrators: LD, NAC-based, rotation-based
- NAC-free methods: LZ, ZN, BLLZ workflow
- State tracking: LD, min-cost, stochastic tracking, F-tracking
- Phase corrections: time-overlap or NAC-based
- Couplings at the KS and TD-DFT levels
- Other: removal of molecular translation/rotation, ML-based methods, TC-NBRA

Akimov, A. V. "Fundamentals of Trajectory-Based Methods for Nonadiabatic Dynamics" Editor(s): Manuel Yáñez, Russell J. Boyd, *Comprehensive Computational Chemistry (First Edition), Elsevier* **2024** Pages 235-272, ISBN 9780128232569, link **Book Chapter in Book: "Comprehensive Computational Chemistry"**

Models Database: Tully, Holstein, spin-Boson, Eckart barriers, SSY, Henon-Heiles, etc.

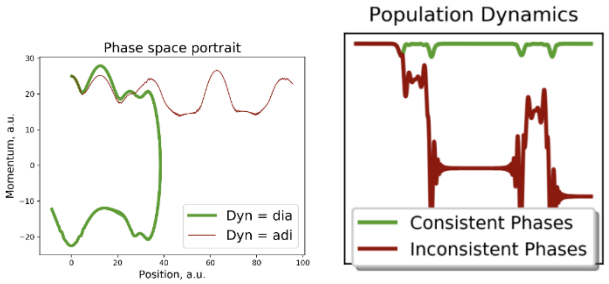
Interfaces: CP2K, MOPAC, OpenMolcas, PySCF, Quantum Espresso, eQE, DFTB+, GAMESS, ErgoSCF, Psi4, …

Some methodological highlights

University at Buffalo
The State University of New York

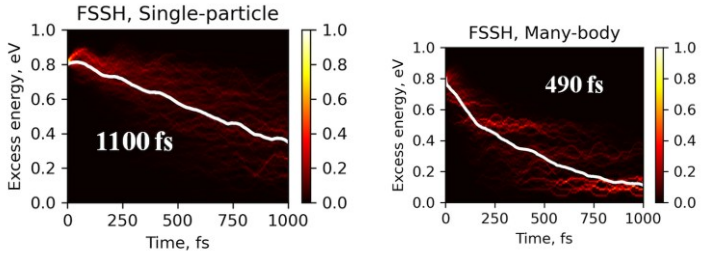
Phase correction for NACs

Akimov *JPCL* **2018** 9, 6096-6102



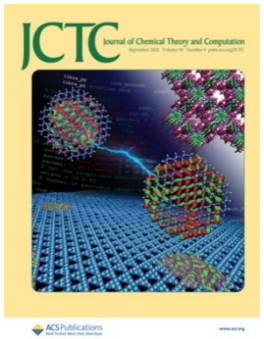
Many-body (TD-DFT) NA-MD

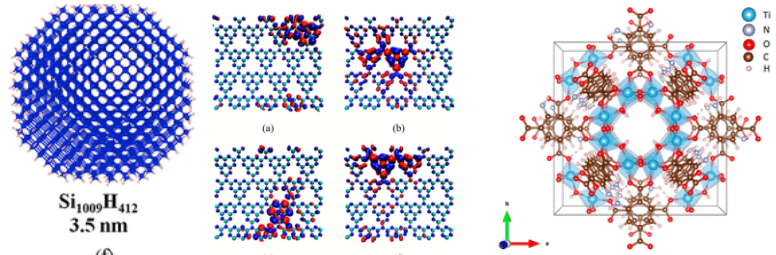
Smith, B.; Shakiba, M.; AVA *JCTC* **2021**, 17, 678
Smith, B.; Shakiba, M.; AVA *JPCL* **2021**, 12, 2444



NA-MD with xTB – applications to large systems

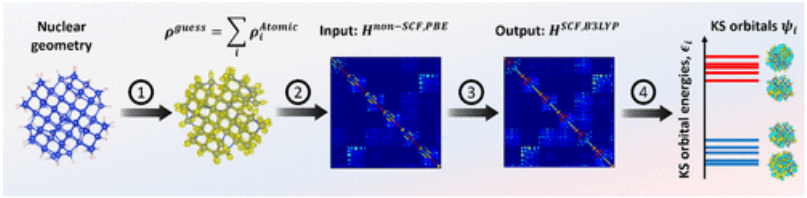
Shakiba, M.; Stippel, E.; Li, W.; Akimov, A. V. *JCTC* **2022** 18, 5157





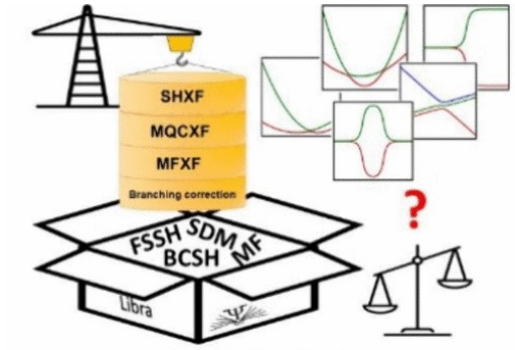
ML for KS Hamiltonian mapping

Shakiba, M.; Akimov, A.V. *JCTC* **2024** 20, 2992



Exact factorization methods

Han, D.; AVA *JCTC* **2024** 20, 5022



Population scores:

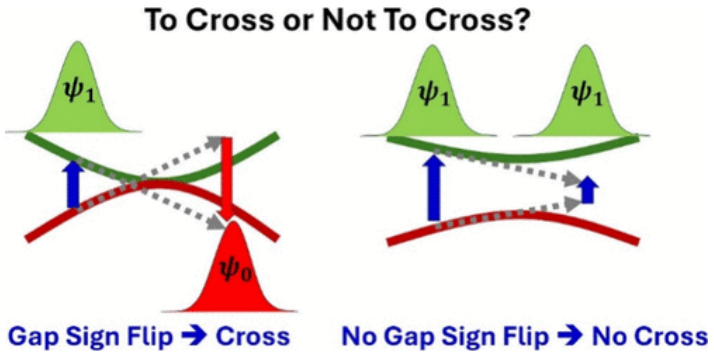
SHXF > MQCXF > BCSH > SDM ≈ MFXF > FSSH ≈ MF

Coherence scores:

BCSH > SHXF > MQCXF > MFXF > SDM > FSSH ≈ MF

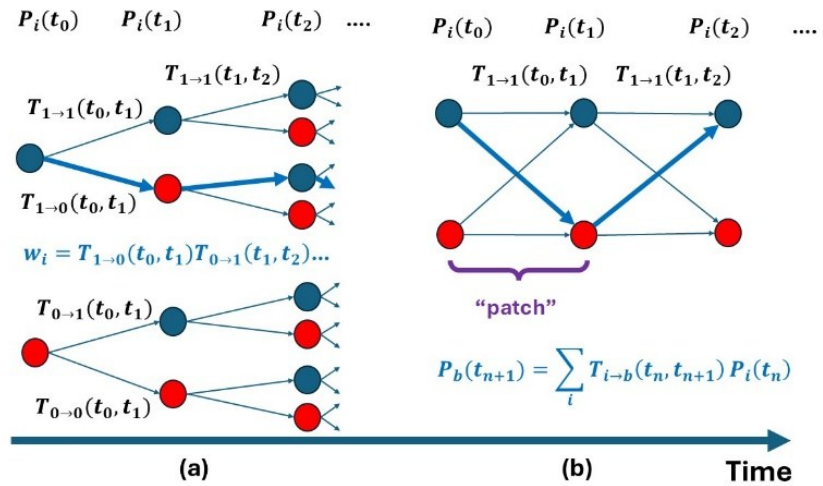
Force-based state tracking

Akimov, V. A. *JPCL* **2024** 15, 11944



FISH: Hop resummation in the NBRA workflow

Han, D.; Shakiba, M.; AVA *JPCL* **2025** 16, 7168



Model Hamiltonians

University at Buffalo
The State University of New York

Model Hamiltonians:

- Tully models I, II, III, generalized — Tully, J. C. *JCP* **1990**, *93*, 1061–1071
- Parandekar-Tully — Parandekar, P. V.; Tully, J. C. *JCP* **2005**, *122*, 094102; Parandekar, P. V.; Tully, J. C. *JCTC* **2006**, *2*, 229–235.
- Dual Rosen-Zener-Demkov
- Dual Landau-Zener-Stuckelberg — Sci. Rep. 6, 24198 (2016); J. Xu and L. Wang JCP 150, 164101 (2019)
- Renner-Teller
- Dumbbell geometry (Subotnik)
- Double arch geometry (Subotnik) — J. E. Subotnik and N. Shenvi JCP 134, 024105 (2011); J. Xu and L. Wang JCP 150, 164101 (2019)
- Shenvi-Subotnik-Yang — Shenvi, N.; Subotnik, J.; Yang, W. JCP 2011, 135, 024101
- Linear Vibronic Coupling (LVC) — Izmaylov, A. F.; Mendive-Tapia, D.; Bearpark, M. J.; Robb, M. A.; Tully, J. C.; Frisch, M. J. JCP, 2011, 135, 234106; Sun, X.; Geva, E. JCP **2016**, 144, 244105

- Generalized LVC (e.g. spin-boson) — Runeson, J. E., Manolopoulos, D. E. *JCP* **2023**, 159, 094115; Tempelaar, R., Reichman, D. R. *JCP* **2018**, 148, 102309; Wang, L., Prezhdo, O. V. *JPCL* **2014**, 5, 713–719; Jain, A., Alguire, E.; Subotnik, J. E. *JCTC* **2016**, 12, 5256–5268; Bondarenko, A. S., Tempelaar, R. *JCP* **2023**, 158, 054117; Mannouch, J. R., Richardson, J. O. *JCP* **2023**, 158, 104111

- Morse models — Corondao, E. A.; Xing, J.; Miller, W. H. Chem. Phys. Lett. 2001, 349, 521-529
- Holstein (many versions) — e.g. Qiu, J.; Bai, X.; Wang, L. *JPCL* **2018**, 9, 4319-4325;
- Henon-Heiles — Sim, E.; Makri, N. *JCP* **1995**, 102, 5616-5625
- Esch-Levine (linear crossings) — Esch, M. P.; Levine, B. G. *JCP* **2020**, 153, 114104;
- Ferretti — Ferretti, A.; Granucci, G.; Lami, A.; Persico, M.; Villani, G. *JCP* **1996**, 104, 5517–5527
- Granucci-Persico (2 models) — Granucci, G., Persico, M. *JCP* **2007**, 126, 134114; Granucci, G., Persico, M.; Zoccante, A. *JCP* **2010**, 133, 134111
- 1D and 2D Eckart barrier (Martens) — L. Wang, C.C. Martens, Y. Zheng, *JCP* **2012**, 137, 34113;
- and more…

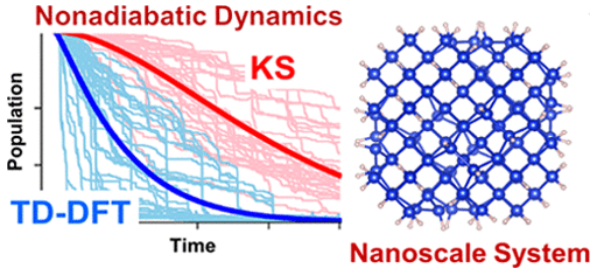
Some application highlights

University at Buffalo
The State University of New York

Clusters and molecules (0D systems)

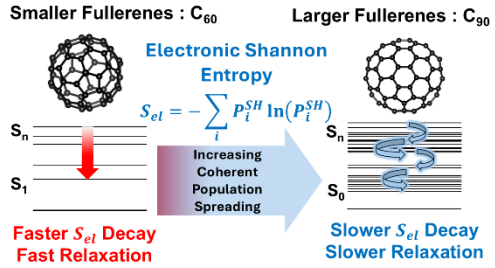
Si and CdSe nanoclusters: TD-DFT vs. KS

Shakiba, M.; Stippel, E.; Li, W.; AVA *JCTC* **2022** 18, 5157



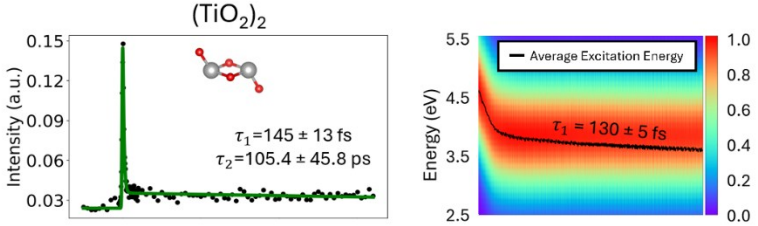
Excited state relaxation in Fullerenes

Yasin, K.; Shakiba, M.; AVA *JPCL* **2026** 17, 2812



Atomically-precise metal oxide clusters

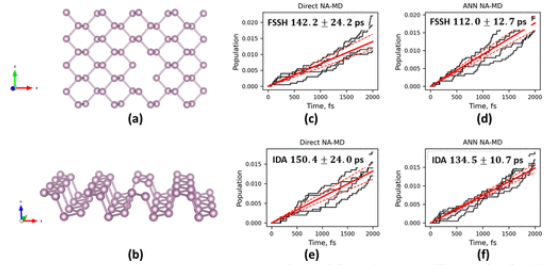
Recio-Poo, M.; Rotteger, C. H.; Illas, F.; Bromley, S. T.; Morales-García, A.; Sayres, S. G.; AVA *JACS* **2025** 147, 40900



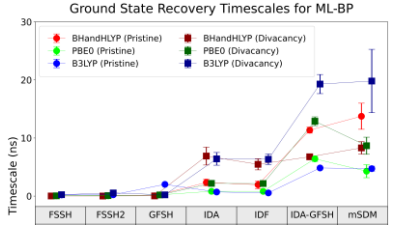
2D systems

Monolayers of black phosphorus

Akimov, A. V. *JPCL* **2021**, *12*, 12119–12128.

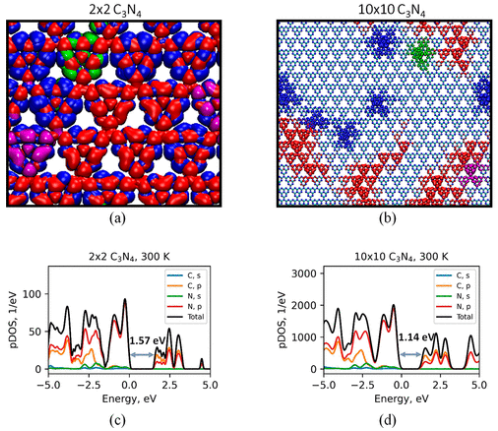


Zabihi, H.; Zhang, Q.; Khayati, G. R.; Irannejad, A.; AVA *JCTC* **2025** 21, 7991



g-C_3N_4 with 6000+ atoms

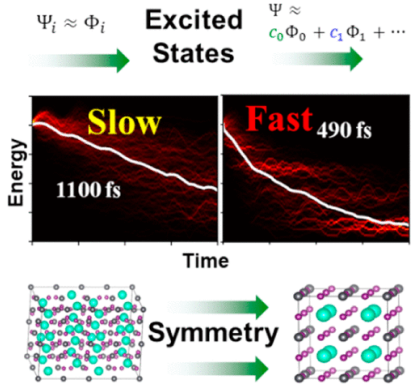
Shakiba, M.; Akimov, A.V. *JPCC* **2023** 127, 9083



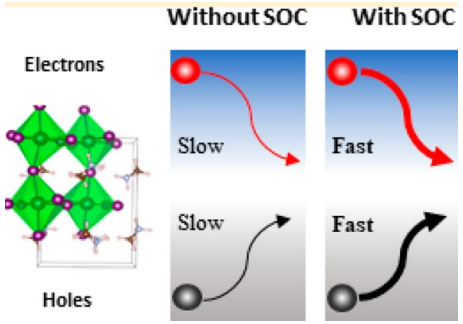
3D systems

Lead halide perovskites: MB and SOC effects

Smith, B.; Shakiba, M.; AVA *JPCL* **2021**, 12, 2444

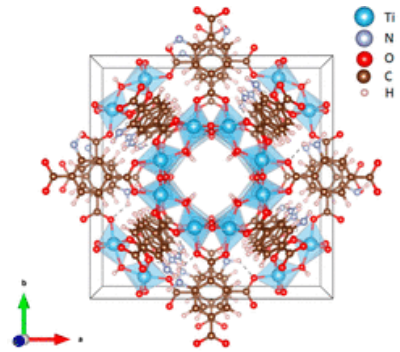


Li, W.; Zhou, L.; Prezhdo, O. V.; AVA *ACS Energy Lett.* **2018**, *3*, 2159–2166.



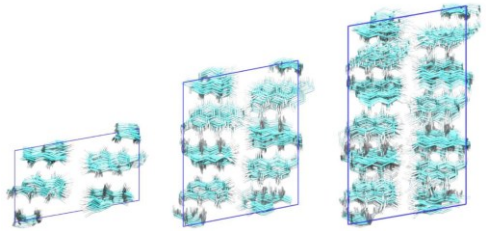
MOFs: With xTB

Shakiba, M.; Stippel, E.; Li, W.; AVA *JCTC* **2022** 18, 5157



Molecular crystalls

Zhang, Q.; Shao, X.; Li, W.; Mi, W.; Pavanello, M.; AVA *JPCM* **2024** 36, 385901



Community Use: Some papers that use Libra

University at Buffalo
The State University of New York

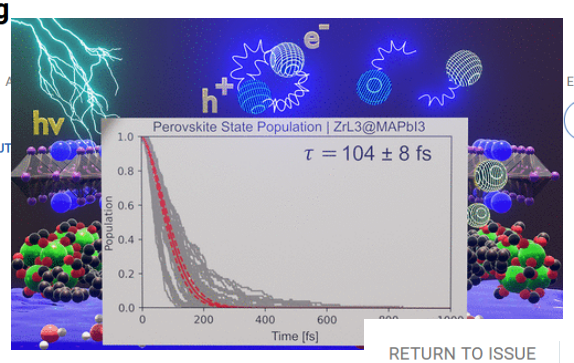
RETURN TO ISSUE | < PREV | **LETTER** | NEXT >

Charge Carrier Dynamics at the Interface of 2D Metal–Organic Frameworks and Hybrid Perovskites for Solar Energy Harvesting

Robert Stanton and Dhara J. Trivedi*

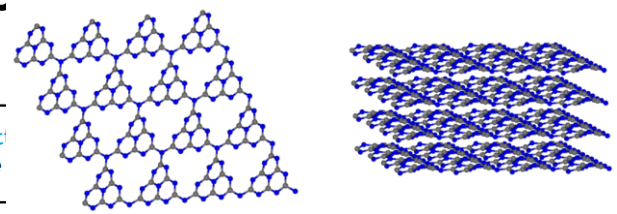
Cite this: *Nano Lett.* 2023, 23, 24, 11932–11939
Publication Date: December 15, 2023
https://doi-org.gate.lib.buffalo.edu/10.1021/acs.nanolett.3c04054
Copyright © 2023 American Chemical Society
Request reuse permissions Subscribed

Article Views 675 LEARN ABOUT



Export RIS

Ab initio quantum dynamics of charge carriers in graphitic carbon nitride nanosheets



Cite as: J. Chem. Phys. **153**, 054701 (2020); ht
Submitted: 13 April 2020 • Accepted: 11 June

Sraddha Agrawal, Wei Lin, Oleg V. Prezhdo, et al.

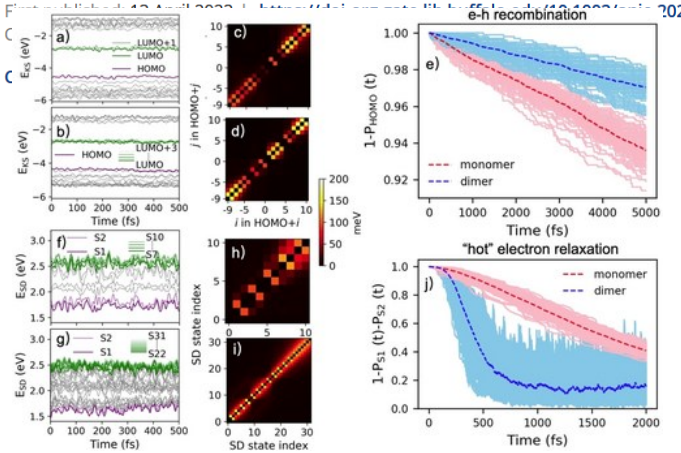
Angewandte International Edition Chemie | GDCh A Journal of the German Chemical Society

Research Article | Full Access

Insight into the Molecular Mechanism for Enhanced Longevity of Supramolecular Vesicular Photocatalysts

Dr. Yannan Liu, Dr. Fulu Zheng, Haojie Dai, Chuanshuang Chen, Yajing Chen, Dr. Haolin Wu, Prof. Chunyang Yu, Prof. Yiyong Mai, Prof. Thomas Frauenheim, Prof. Yongfeng Zhou

First published: 12 April 2023 | https://doi-org.gate.lib.buffalo.edu/10.1002/anie.202302126 |



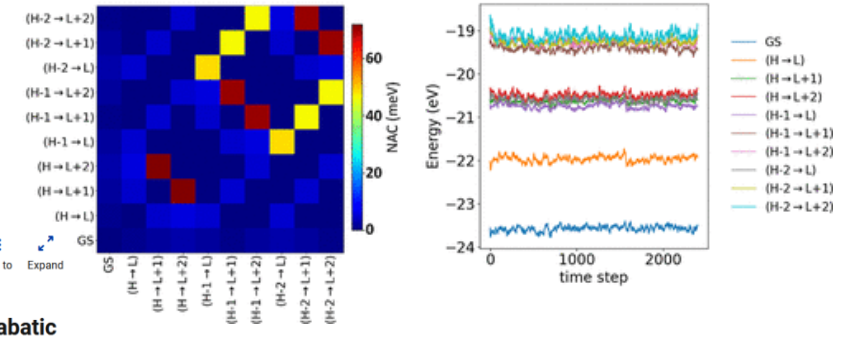
RETURN TO ISSUE | < PREV | **A: NEW TOOLS AND MET...** | NEXT >

Convergence of Time-Derivative Nonadiabatic Couplings in Plane-Wave DFT Calculations

Alva D. Dillon and Rebecca L. M. Gieseking*

Cite this: *J. Phys. Chem. A* 2023, 127, 45, 9612–9620
Publication Date: November 4, 2023
https://doi-org.gate.lib.buffalo.edu/10.1021/acs.jpca.3c04858
Copyright © 2023 American Chemical Society
Request reuse permissions Subscribed

Article Views 196 | Altmetric - | Citations 1
LEARN ABOUT THESE METRICS



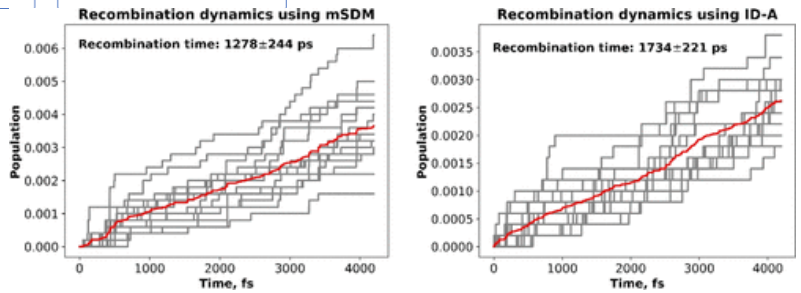
The Journal of Physical Chemistry C > ASAP > Article
Subscribed
Cite Share Jump to Expand

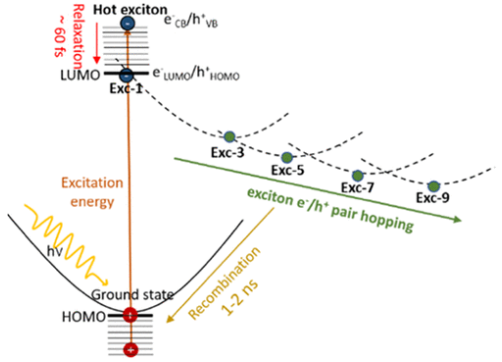
C: ENERGY CONVERSION AND STORAGE | August 9, 2024

Excitons in Hematite Fe_2O_3: Short-Time Dynamics from TD-DFT and Non-Adiabatic Dynamics Theories

Lili Rassouli, Mohammad Shakiba, Alexey V. Akimov, Xuyan Ma, and Michel Dupuis*

Open PDF | Supporting Information (1)





Community

University at Buffalo
The State University of New York

Mohammad Shakiba



Brendan Smith



Daeho Han



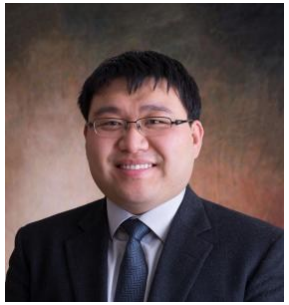
Sophya Garashchuk



Amber Jain



Xiang Sun



Thomas Niehaus



- Taras Khvorost
- Leonardo Araujo
- Joseph Kelly
- Victor Suarez
- Xuecheng Shao
- Wenhui Mi
- Matthew Dutra
- Alva Dillon
- Miguel Recio Poo
- Qingxin Zhang
- Kevin Walsh
- Kosuke Sato
- Ekadashi Pradhan
- Dhara Trivedi
- Wei Li
- Story Temen
- Rebecca Gieseking
- Michele Pavanello
- Michele Dupuis
- Lili Rassouli
- Xuyan Ma
- Kosar Yasin
- more coming…

Published March 22, 2026 | Version v5.12.0

Libra

Akimov, Alexey V.[1]; Shakiba, Mohammad[1]; Han, Daeho[1]; Smith, Brendan[1]; Dutra, Matthew[2]; Sato, Kosuke[3]; Suarez, Victor[4]; Temen, Story[1]; Wang, Yuchen[5]; Li, Wei[6]; Khvorost, Taras[7]; Yasin, Kosar[1]; Sun, Xiang[8]; Balducci, Simone[9]; Zhang, Qingxin[1]; Niehaus, Thomas[10]; Stippell, Elizabeth[11]; Gerasimov, Igor S.[12]; Amber, Jain[13]

- Several new collaborations are ongoing
- Open for more …

Training Resources

University at Buffalo
The State University of New York

The first one – Winter 2022:



Libra Winter School on Excited States and Nona Materials 2022

Libra Workshop and Summer School on Excited States and Nonadiabatic Dynamics 2024

https://compchem-cybertraining.github.io/Libra_Winter_School_2022/

https://compchem-cybertraining.github.io/Libra_Summer_School_2024/

Join Slack: https://join.slack.com/t/quantumdynamicshub/shared_invite/zt-mjbhjssx-GGhsbYHxeBMvhmumK_j7LA

The second one – Summer 2024:

Comutational Chemistry CyberTraining

https://github.com/compchem-cybertraining

Learning Libra: The dedicated set of tutorials: https://github.com/compchem-cybertraining/Tutorials_Libra

Learning Libra AND theory AND other packages: CyberTraining workshops

- 2026: https://compchem-cybertraining.github.io/Cyber_Training_Workshop_2026/
- 2023: https://compchem-cybertraining.github.io/Cyber_Training_Workshop_2023/
- https://github.com/compchem-cybertraining/Cyber_Training_Workshop2023_course_projects
- 2022: https://compchem-cybertraining.github.io/Cyber_Training_Workshop_2022/
- 2021: https://compchem-cybertraining.github.io/Cyber_Training_Workshop_2021/





Excited States and Nonadiabatic Dynamics CyberTraining Workshop 2023

This workshop: https://github.com/compchem-cybertraining/Libra_CP2K_autumn_school_2026/tree/main

University at Buffalo
The State University of New York

Part 2: Theory Overview and Hands On

A lot more details: https://compchem-cybertraining.github.io/Cyber_Training_Workshop_2026/_episodes/04-tsh

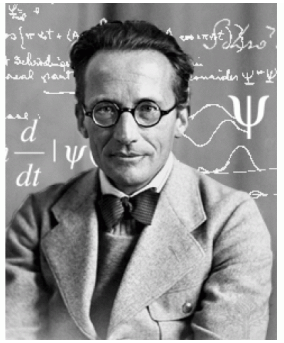
Adiabatic and Nonadiabatic Dynamics

University at Buffalo
The State University of New York

$$i\hbar \frac{\partial \Psi(t,\boldsymbol{r},\boldsymbol{R})}{\partial t} = H(t,\hat{\boldsymbol{r}},\hat{\boldsymbol{R}})\Psi(t,\boldsymbol{r},\boldsymbol{R})$$

$$\Psi(t,\boldsymbol{r},\boldsymbol{R}) = \sum_i \chi_i(t,\boldsymbol{R})\Phi_i(\mathbf{r};\mathbf{R}(t))$$

Except for in the exact factorization ansatz!



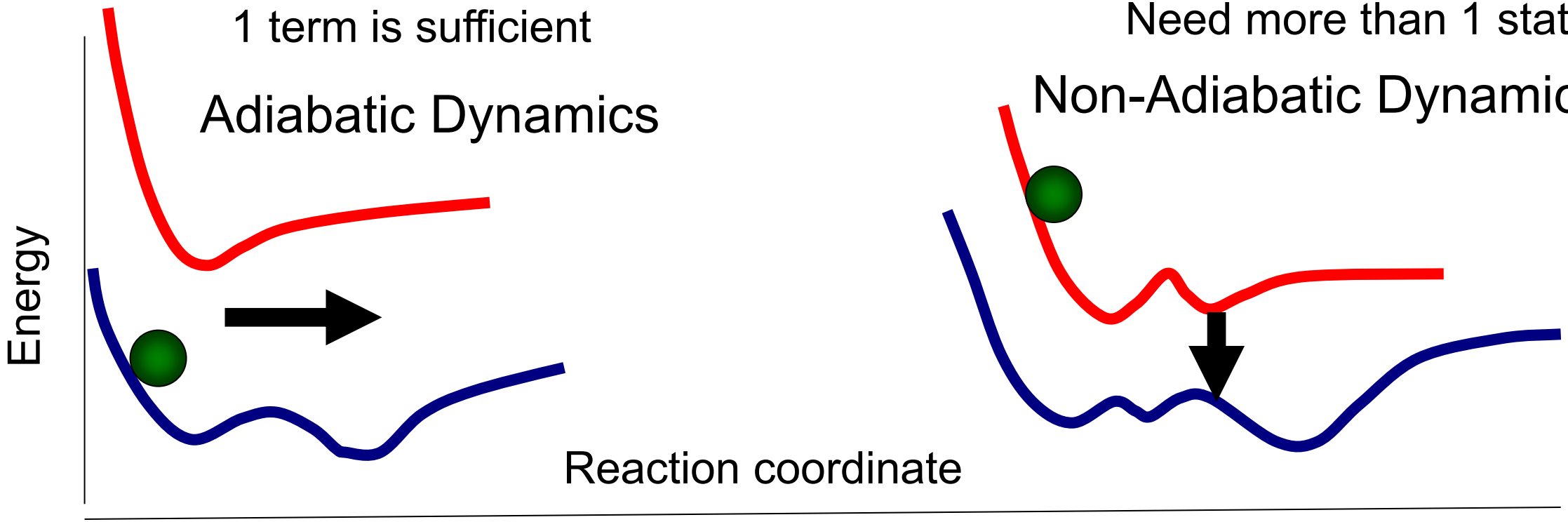
Adiabatic approximation: keep only one term

1 term is sufficient

Adiabatic Dynamics

Need more than 1 state

Non-Adiabatic Dynamics



Classical Path Approximation

University at Buffalo
The State University of New York

$$i\hbar \frac{\partial \Psi(t, \boldsymbol{r}; \boldsymbol{R}(t))}{\partial t} = H(t, \hat{\boldsymbol{r}}; \boldsymbol{R}(t)) \Psi(t, \boldsymbol{r}; \boldsymbol{R}(t))$$

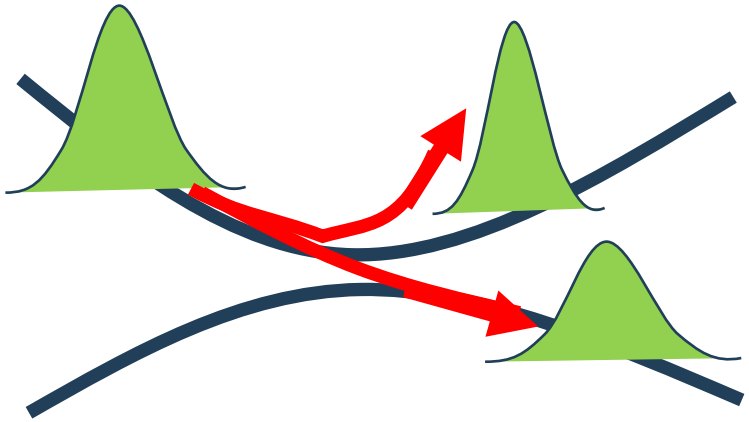
$$\Psi(t, \boldsymbol{r}; \boldsymbol{R}(t)) = \sum_i c_i(t) \Phi_i(\boldsymbol{r}; \mathbf{R}(\boldsymbol{t}))$$

$$i\hbar \frac{\partial}{\partial t} c_i = E_i(\boldsymbol{R}(t)) c_i - i\hbar \sum_j \left\langle \Phi_i \middle| \frac{\partial}{\partial t} \middle| \Phi_j \right\rangle c_j$$

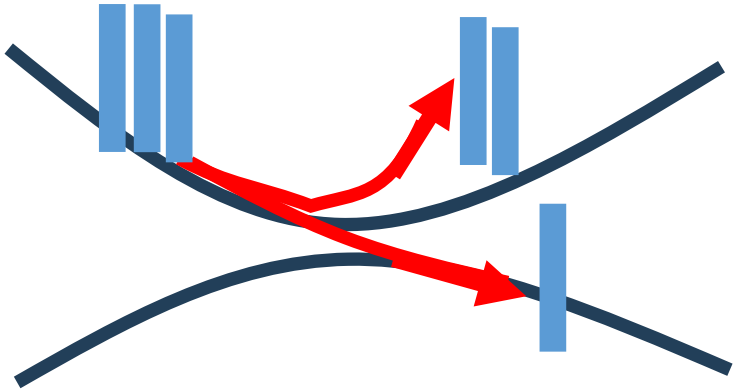
$c_i(t)$ – state amplitudes (electronic coordinates)

$$\dot{\boldsymbol{R}}_\alpha(t) = \frac{\boldsymbol{P}_\alpha}{M_\alpha} \qquad \dot{\boldsymbol{P}}_\alpha(t) = \boldsymbol{F}_{\alpha,i}$$

$\{\boldsymbol{R}, \boldsymbol{P}\}$ - nuclear coordinates and momenta



Quantum Nuclei: Wavepackets



Classical Trajectories

The general NA-MD workflow

University at Buffalo
The State University of New York

Initialization

Nuclear dynamics

step 1

$$\dot{p}_i = -\frac{\partial H}{\partial r_i} \quad \dot{r}_i = \frac{\partial H}{\partial p_i}$$

Stationary adiabatic states

step 2

$$\hat{H}_{el}\psi_i = E_i\psi_i$$

analytical models or electronic structure codes

Non-adiabatic Couplings

$$d_{ij} = \frac{\langle \psi_i(t) | \psi_j(t+dt) \rangle - \langle \psi_i(t+dt) | \psi_j(t) \rangle}{2dt}$$

step 3

Integrate Coherent Electronic Dynamics

$$\Psi(\mathrm{r},\mathrm{R},\mathrm{t}) = \sum_i c_i(t)\psi_\mathrm{i}(\mathrm{r};\mathrm{R}(\mathrm{t})) \qquad i\hbar\frac{\partial \mathrm{c_i(t)}}{\partial t} = \sum_j \left(E_i\delta_{ij} - i\hbar d_{ij}\right)c_j$$

NBRA

Decoherence 1

$$c_i \rightarrow c_i \exp\left(-\frac{\Delta t}{\tau_{ij}}\right), \forall i \neq j$$

as in SDM

Proposed Hops Decoherence 2

$$P_{i\rightarrow j} = \Delta t * Re\left(\frac{2\frac{p}{m}d_{ij}c_i^* c_j}{c_i^* c_i}\right)$$

or as in DISH

step 4

Accept Hops

based on energy conservation or

$$P_{i\rightarrow f}^A = \min\left(1, exp\left(-\frac{\Delta E}{k_B T}\right)\right)$$

Change of state/Velocity rescaling

change active electronic state, rescale velocity

NAC: Hammes-Schiffer, S.; Tully, J. C. *JCP* **1994**, *101*, 4657
Phase tracking: Akimov, A. V. *JPCL* **2018** 9, 6096

Local Diabatization: Granucci, G.; Persico, M.; Toniolo, A. *JCP* **2001**, *114*, 10608; Plasser, F.; Granucci, G.; Pittner, J. et al. *JCP* **2012**, 137, 22A514; Shakiba, M.; Akimov, A. V. *TCA* **2023**, 142, 68

SDM: Granucci, G.; Persico, M. *JCP* **2007**, *126*, 134114;
mSDM: Smith, B. A.; Akimov, A. V. *JCP*, **2019**, 151, 124107;
IDA: Nelson, T.; Fernandez-Alberti, S.; Roitberg, A. E.; Tretiak, S. *JCP* **2013**, *138*, 224111.

FSSH: Tully, J. C. *JCP*, **1990**, 93, 1061

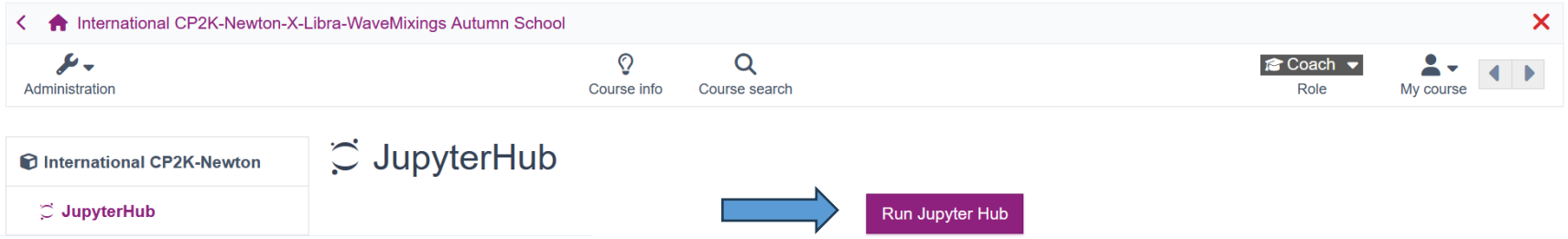
NBRA: Prezhdo, O. V.; Duncan, W. R.; Prezhdo, V. V. *Prog. Surf. Sci.* **2009**, *84*, 30

Getting Started on NESH

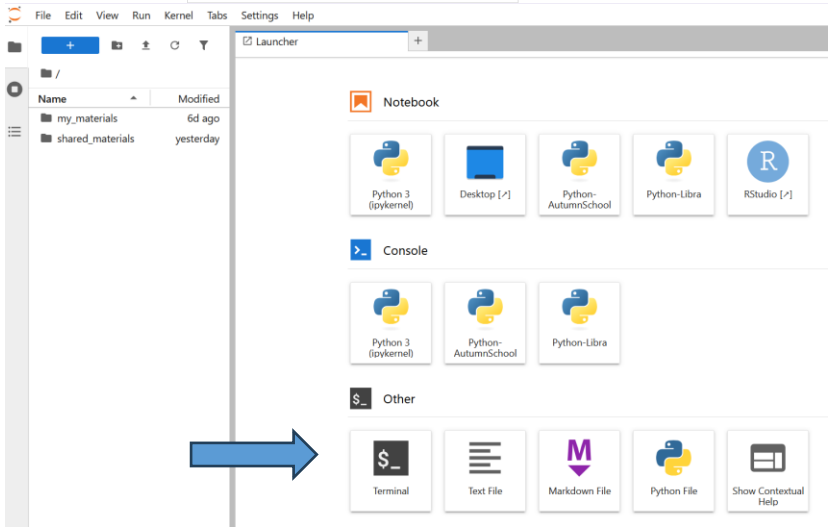
University at Buffalo
The State University of New York

1. Log in to the OpenOlat and run JupyterHub

https://lms.uni-kiel.de/url/RepositoryEntry/5837357358



2. Open Terminal



3. SSH to NESH

ssh -X xeext951@nesh-login.rz.uni-kiel.de

Use your username

4. Edit your .bashrc

xeext951@nesh-login3:~

```
# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
        . /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature:
# export SYSTEMD_PAGER=

# User specific aliases and functions

eval "$(/gxfs_home/cau/xeext951/Conda/bin/conda shell.bash hook)"
module use /gxfs_home/cau/xeext951/MODULES
module load libra/devel
```

```
eval "$(/gxfs_home/cau/xeext951/Conda/bin/conda shell.bash hook)“
module use /gxfs_home/cau/xeext951/MODULES
module load libra/devel
```

5. Restart terminal

Getting Started on NESH

University at Buffalo
The State University of New York

1. Clone tutorial repository

```
>> cd
>> git clone https://github.com/compchem-cybertraining/Libra_CP2K_autumn_school_2026.git
```

2. Conduct calculations

3. Copy files back to JupyterHub for visualization

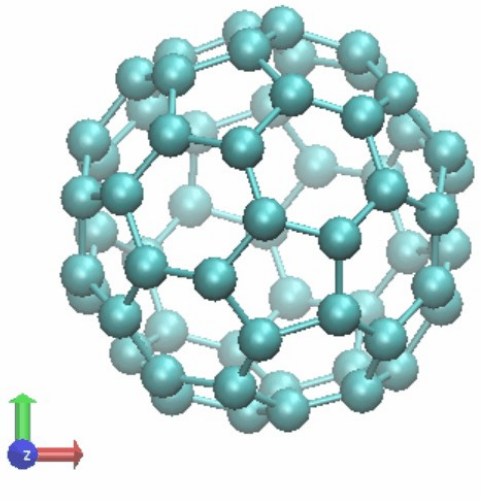
```
>> scp xeext951@nesh-login.rz.uni-kiel.de:/gxfs_home/cau/xeext951/Libra_CP2K_autumn_school_2026/1_step1/out_MgC60.log .
```

Step 1: Molecular Dynamics and MD Alignment

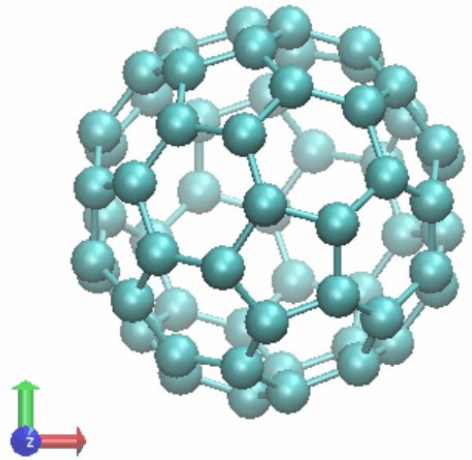
University at Buffalo
The State University of New York

Nuclear Dynamics – Classical:

$$\dot{R}_k = \frac{P_k}{M_k} \qquad \dot{P}_k = -\nabla_{R_k} E_i$$



C_{60}
Initial Trajectory



C_{60}
Aligned Trajectory



Kosar Yasin

Eliminate translational and rotational degrees of freedom.

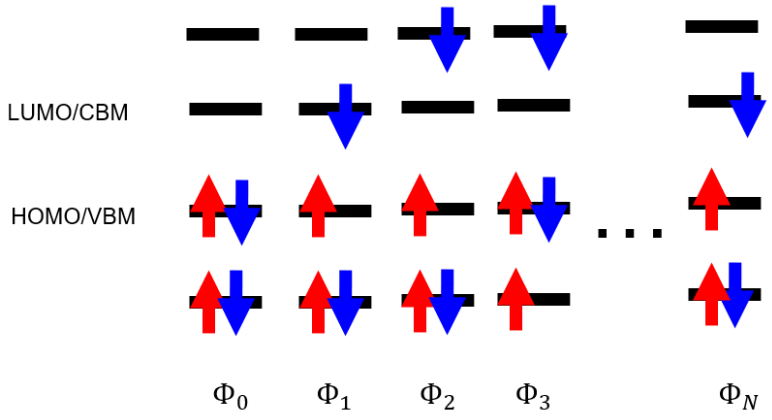
Translational & Rotational motion may make $S_{ii}(t, t')$ small and some $S_{ij}(t, t')$ accidentally large leading spurious increased coupling between physically uncoupled states, which can contribute to abrupt hops

>> cd 1_step1
>> sbatch submit.slm

Steps 2 & 3: Computing time-overlaps and NACs

University at Buffalo
The State University of New York

Single-particle description of excited states



Single-particle states $\Psi_i \approx \Phi_i$

- Usually okay for lower excited states
- For less symmetric systems

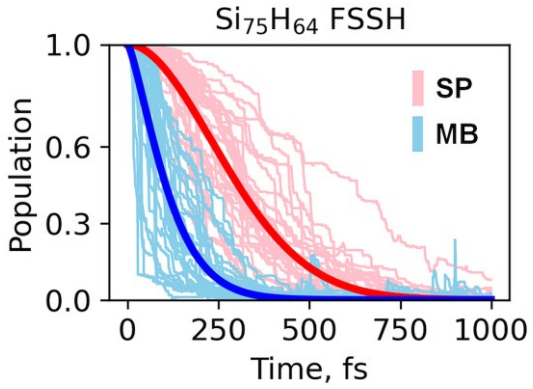
da Silva Oliboni, R.; et al *JPCC* **2016**, *120*, 27688

Akimov, A. V.; Prezhdo, O. V. *JCTC* **2013**, *9*, 4959

Oberhofer, H.; Blumberger, J. *PCCP* **2012**, *14*, 13846

Many-body description generally accelerates the dynamics

$$\Psi = c_0\Phi_0 + c_1\Phi_1 + c_2\Phi_2 + \cdots$$



Smith, B. A.; Shakiba, M.; AVA *JCTC*, **2021**, 17, 678

Smith, B. A.; Shakiba, M.; AVA *JPCL* **2021,** 12, 2444

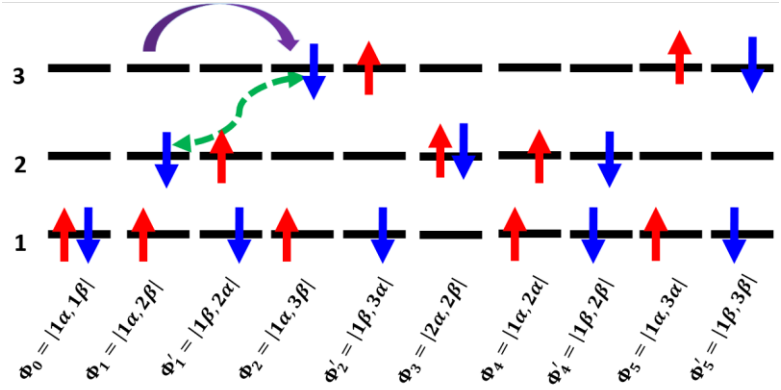


Brendan Smith



Mohammad Shakiba

Spin-adapted configurations



$$S_0 = \Phi_0 \qquad S_1 = \frac{1}{\sqrt{2}}(\Phi_1 - \Phi_1') \qquad S_2 = \frac{1}{\sqrt{2}}(\Phi_2 - \Phi_2')$$

	Φ_0	Φ_1	Φ_1'	Φ_2	Φ_2'	Φ_3	Φ_4	Φ_4'	Φ_5	Φ_5'
Φ_0	0	d_{12}	$-d_{12}$	d_{13}	$-d_{13}$	0	0	0	0	0
Φ_1		0	0	d_{23}	0	d_{12}	0	0	0	0
Φ_1'			0	0	d_{23}	$-d_{12}$	0	0	0	0
Φ_2				0	0	0	0	0	0	0
Φ_2'					0	0	0	0	0	0
Φ_3						0	0	0	0	0
Φ_4							0	0	0	0
Φ_4'								0	0	0
Φ_5									0	0
Φ_5'										0

	S_0	S_1	S_2
S_0	0	$\sqrt{2}d_{12}$	$\sqrt{2}d_{12}$
S_1		0	d_{23}
S_2			0

Shakiba, M.; Stippel, E.; Li, W.; AVA *JCTC* **2022,** 18, 5157

Steps 2 & 3: Computing time-overlaps and NACs

University at Buffalo
The State University of New York

Step 2: MO (KS) time-overlaps

$$\psi^{KS}_{\boldsymbol{k},i}(t) = \sum_j U_{\boldsymbol{k},ji}(t)\beta_{\boldsymbol{k},j}(t) \qquad \beta_{\boldsymbol{k},a}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{t}} e^{i\boldsymbol{k}\boldsymbol{t}}\phi_a(\boldsymbol{r}-\boldsymbol{t})$$

$$St^B_{\boldsymbol{k},a,\boldsymbol{k}',b} = \langle\beta_{\boldsymbol{k},a}|\beta_{\boldsymbol{k}',b}\rangle = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}'\boldsymbol{R}}\frac{1}{N}\sum_{\boldsymbol{t}} e^{i(\boldsymbol{k}'-\boldsymbol{k})\boldsymbol{t}} St^{AO}_{ab}(\boldsymbol{R})$$

Same k-point $St^B_{\boldsymbol{k},a,\boldsymbol{k},b} = \langle\beta_{\boldsymbol{k},a}|\beta_{\boldsymbol{k},b}\rangle = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} S_{ab}(\boldsymbol{R})$

Gamma k-point $St^B_{\boldsymbol{k},a,\boldsymbol{k},b} = \langle\beta_{\boldsymbol{k},a}|\beta_{\boldsymbol{k},b}\rangle = \langle\phi_a|\phi_b\rangle = St^{AO}_{ab}$

Smith, B. A.; Shakiba, M.; AVA *JCTC*, **2021**, 17, 678
Smith, B. A.; Shakiba, M.; AVA *JPCL* **2021,** 12, 2444

} via "cube" files inefficient

Current:

- numpy.muptiprocessing
- Libint2; using *molden* format
- xTB implementation if CP2k

Shakiba, M.; Stippel, E.; Li, W.; AVA *JCTC* **2022,** 18, 5157



Mohammad Shakiba

https://github.com/compchem-cybertraining/Tutorials_Libra/blob/master/14_molecular_integrals/4_libint2_advanced

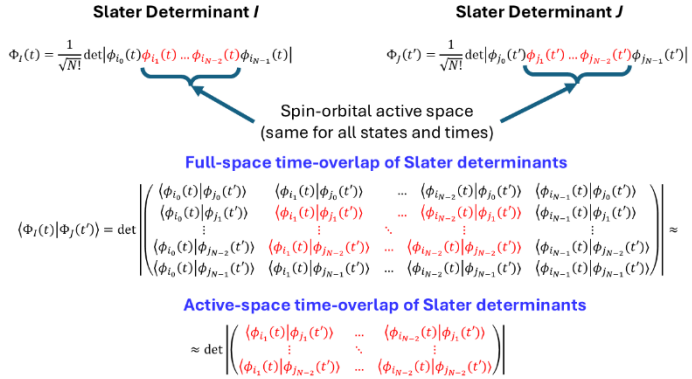
Step 3: TD-DFT time-overlaps

Canonical ordering: $(-i) < (+i) < (-j) < (+j) \quad \text{for} \quad i < j,$

Slater Determinant time-overlaps from KS time-overlap

$$\langle SD_I(t)|SD_J(t')\rangle = \det|St(t,t')^B|$$

Can use reduced number of electrons



Spin adaptation $|\boldsymbol{CSF}\rangle = |\mathbf{SD}\rangle\boldsymbol{L}$

TD-DFT pseudowavefunction: Usually called **X**

$$\psi_{I,TD-DTF} = \sum_{i\in occ, a\in virt} t_{i\to a}\mathrm{CSF}_{i\to a} \leftrightarrow |\boldsymbol{\psi_{TD-DFT}}\rangle = |\boldsymbol{CSF}\rangle\boldsymbol{T}$$

$$\boldsymbol{St} = \langle\boldsymbol{\psi}(t)|\boldsymbol{\psi}(t')\rangle$$

Lowdin orthogonalization:

$$\boldsymbol{St}(t,t') \to \boldsymbol{S}^{-1/2}(t)\boldsymbol{St}(t,t')\boldsymbol{S}^{-1/2}(t')$$

https://github.com/compchem-cybertraining/Tutorials_Libra/tree/master/15_citools

Step 2: Hands On

University at Buffalo
The State University of New York

"distribute_jobs.py"

```
import os
import sys
import libra_py.packages.cp2k.methods as CP2K_methods


run_slurm = True
submit_template = 'submit_template.slm'
run_python_file = 'run_template.py'
istep = 0
fstep = 20 #1402
njobs = 4
submission_exe = 'sbatch'
# Removing the previous folders if existed. You can keep t
# but Libra will overwrite some of the data if their names
#os.system('rm -rf res job* all_*')

print('Distributing jobs...')
CP2K_methods.distribute_cp2k_libint_jobs(submit_template,
```

- defines how to organize calculations
- creates submission folder for each job
- submits the jobs

"submit_template.slm"

```
#!/bin/bash -l
#SBATCH --account=cyberwksp21
##SBATCH --partition=valhalla  --qos=valhalla
##SBATCH --clusters=faculty
#SBATCH --partition=general-compute  --qos=general-comput
#SBATCH --clusters=ub-hpc
##SBATCH --partition=scavenger  --qos=scavenger
#SBATCH --time=1:00:00
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=12
#SBATCH --cpus-per-task=1
#SBATCH --mem=64000
###SBATCH -C CPU-Gold-6130
###SBATCH --mail-user=mshakiba@buffalo.edu
echo "SLURM_JOBID="$SLURM_JOBID
echo "SLURM_JOB_NODELIST="$SLURM_JOB_NODELIST
echo "SLURM_NNODES="$SLURM_NNODES
echo "SLURMTMPDIR="$SLURMTMPDIR
echo "working directory="$SLURM_SUBMIT_DIR
module use /projects/academic/cyberwksp21/MODULES
module purge
module load cp2k/v25/local

# These two lines are as before
module load libra_ava/devel
python run.py
```

- defines how run each job
- set the HPC resources to use
- Becomes "**submit.slm**" in each job's directory

"run_template.py"

```
# The AO overlaps in spherical or Cartesian coordinates
params['is_spherical'] =  True
# Remove the molden files, which are large files for some systems,
# after the computaion is done for tha system
params['remove_molden'] = True

# Cube visualization using VMD
# For the TiO2 unit cell we do not visualize the cube files,
params['cube_visualization'] = False
if params['cube_visualization']:
    # The only parts that we will change in this template are loading the cu
    params['vmd_input_template'] = '../vmd_cube_template.tcl'
    params['states_to_plot'] = [24,25]
    params['plot_phase_corrected'] = True
    params['vmd_exe'] = 'vmd'
    params['tachyon_exe'] = '/util/academic/vmd/1.9.2/lib/vmd/tachyon_LINUXA
    params['x_pixels'] = 1024
    params['y_pixels'] = 1024
    params['image_format'] = 'bmp'
    params['remove_cube'] = True
    params['all_images'] = path + '/../all_images'

# The results are stored in this folder
params['res_dir'] = path + '/../res'
params['all_pdosfiles'] = path + '/../all_pdosfiles'
params['all_logfiles'] = path + '/../all_logfiles'
# If the xTB calculations are needed, we need an OT procedure
params['cp2k_ot_input_template'] = path + '/../es_ot_temp.inp'
params['cp2k_diag_input_template'] = path + '/../es_diag_temp.inp'
# The trajectory xyz file path
# Note that since it will be run in one of the jobs folders
# we need to put one more .. so that it can recognize the file
params['trajectory_xyz_filename'] = path + '/../Rutile_TiO2_MD-pos-1.xyz'


step2.run_cp2k_libint_step2(params)
```

- defines how to compute time-overlaps
- sets system-specific parameters
- becomes "**run.py**" in each job's directory

"es_diag_temp.inp"

```
&GLOBAL
  PRINT_LEVEL  MEDIUM
  PROJECT_NAME Rutile_TiO2_ENERGY
  RUN_TYPE  ENERGY
&END GLOBAL
&FORCE_EVAL
 METHOD  QS
 &PROPERTIES
   &TDDFPT
      NSTATES     10             # number of excited states
      MAX_ITER    200            # maximum number of Davidson iterations
      CONVERGENCE [eV] 1.0e-5    # convergence on maximum energy change
      &MGRID
         NGRIDS 25
         CUTOFF 100
      &END

      !RESTART     .TRUE.
      !WFN_RESTART_FILE_NAME RESTART.tdwfn

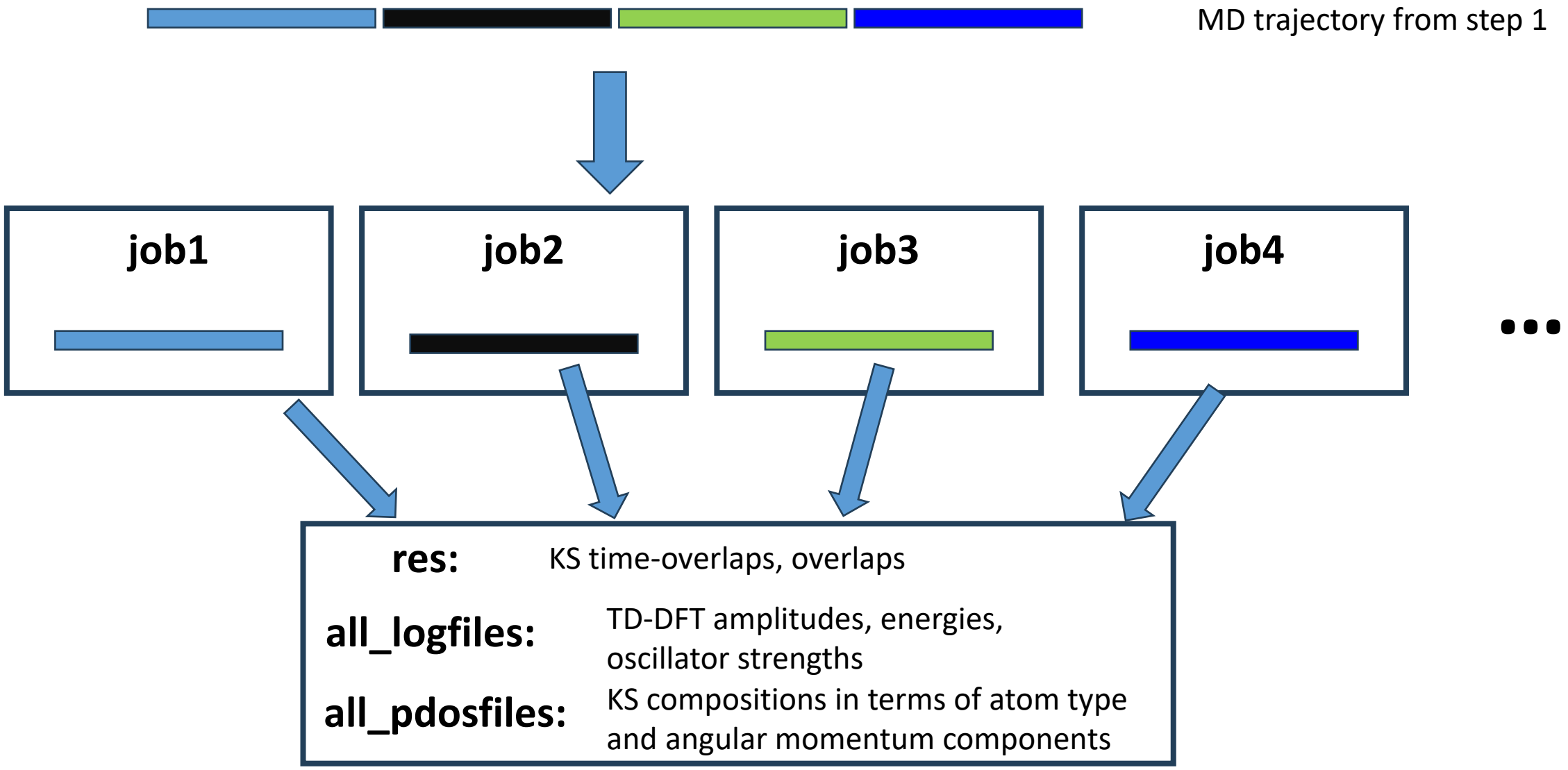
   &END TDDFPT
 &END PROPERTIES
 &DFT
   UKS .FALSE.
   BASIS_SET_FILE_NAME BASIS_MOLOPT
   POTENTIAL_FILE_NAME POTENTIAL
   &MGRID
     NGRIDS 16
     CUTOFF 500
     REL_CUTOFF 50
   &END MGRID
   &QS
     METHOD GPW
     EPS_DEFAULT 1.000E-14
```

- defines parameters for electronic structure calculations with CP2K

>> cd 2_step2
>> python align.py
>> python distribute_jobs.py

Step 2: What happens

University at Buffalo
The State University of New York

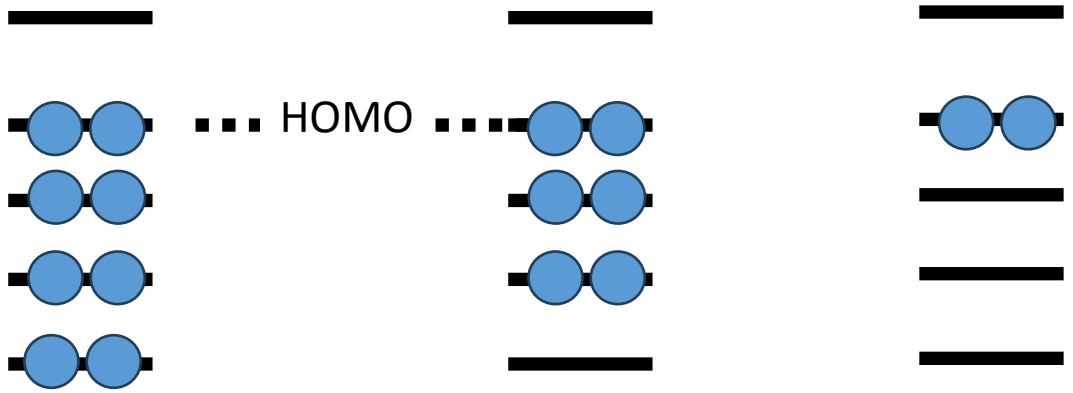


Step 3: Hands On

University at Buffalo
The State University of New York

```
>> cd 3_step3
>> python run.py
```

```
params = {"atom_labels":labels, "timestep":0,
          "dt":1.0*units.fs2au,
          "nelec_act_space":None,
          "ci_threshold":0.001,
          "lowest_orbital": 1,
          "nstates": 11,
          "is_first_time":{0:True},
          "act_state":{0:1},
         }
```



nelec_act_space = None

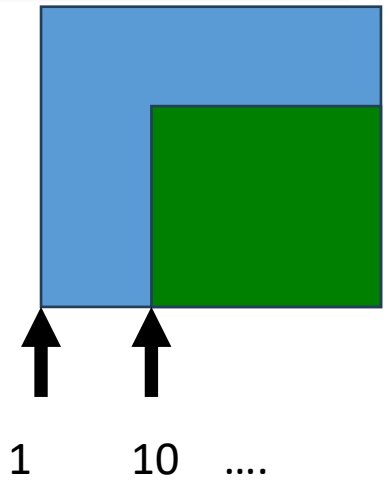
nelec_act_space = 8

nelec_act_space = 6

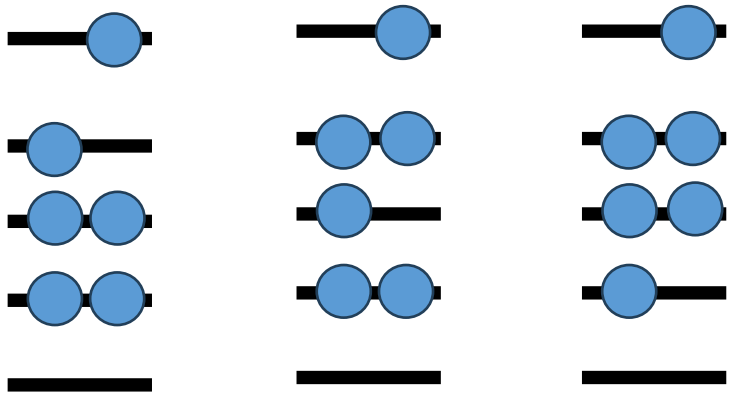
nelec_act_space = 2

In step 2, we use all KS orbitals:

- Big matrix
- ***lowest_orbital*** = 1



Retain configurations such that: $\left|t_{ij}\right|^2$ >ci_threshold



In step 2, we truncate some KS orbitals:

- Smaller matrix
- ***lowest_orbital*** = 10 (example)



- Smaller ***ci_threshold*** requires larger ***nelec_act_space***

Step 3 plotting: Energy vs time, DOS, and UV-VIS

University at Buffalo
The State University of New York

>> python plotting.py

```
# ================== Settings to edit before running ====================
DATA_DIRECTORY = "ci_tio2"
OUTPUT_DIRECTORY = "plots"
ISTEP, FSTEP = 0, 20              # end excluded
DT = 1.0                          # frame spacing, fs (matrices and logs)
# Comment out entries to disable analyses; no CP2K logs are needed unless
# "excitation" or "uvvis" is selected.
PLOTS = [
    "energies",
    "gaps",
    "nacs",
    "overlaps",
    "spectra",
    "excitation",
    "uvvis",
]
# True for the aligned run.py recipe: first output is cache initializati
SKIP_INITIAL_NAC = True
NAC_PAIRS = "all"                 # "all" unique pairs or "adjacent"; one d
UVVIS_FWHM_EV = 0.10              # full width at half maximum, eV
UVVIS_GRID_STEP_EV = 0.005
UVVIS_STATES = None               # None: all EXC_NSTATES; or e.g. [1, 2, 
                                  # excited-state numbers, starting at 1

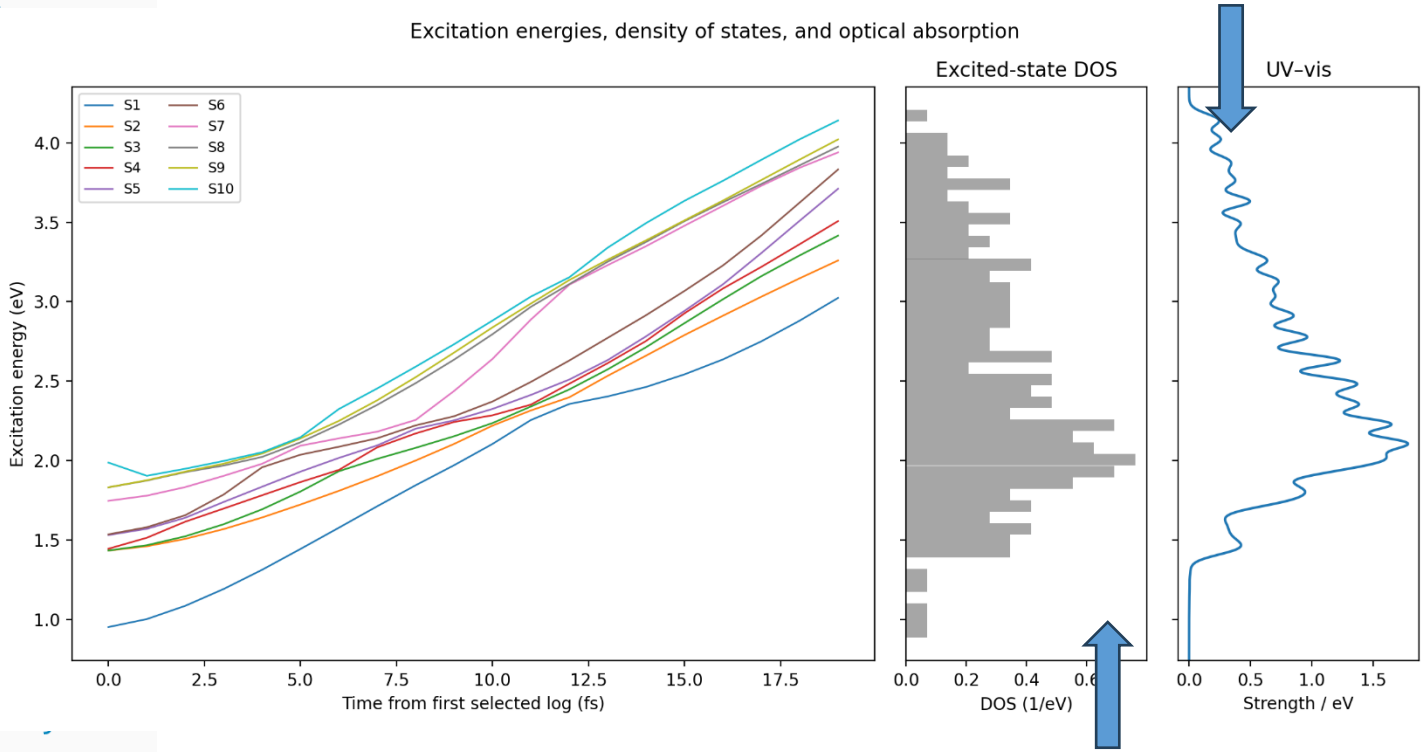
DOS_BINS = 50

# Gap spectra use the saved Step 3 Hamiltonians.
SPECTRUM_PAIRS = [(0, 1)]
SPECTRUM_TEMPERATURE = 300.0      # K, for classical-to-harmonic bath con
SPECTRUM_MAX_WAVENUMBER = 4000.0 # cm^-1, display limit only

# CP2K log settings: select the same trajectory and state set as Step 3.
EXC_LOG_DIRECTORY = "all_logfiles"        # relative to the working dire
EXC_LOG_PATTERN = "step_{step}.log"
EXC_NSTATES = 10                  # excited states only, excludes S0
EXC_CI_THRESHOLD = 0.001          # squared-amplitude cutoff
EXC_IS_UKS = False
EXC_NSDS = 4                      # leading entries/ranks to display
EXC_TOP_TRANSITIONS = 12          # most important orbital identities per state
```

$$G_\sigma(x) = \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma},$$

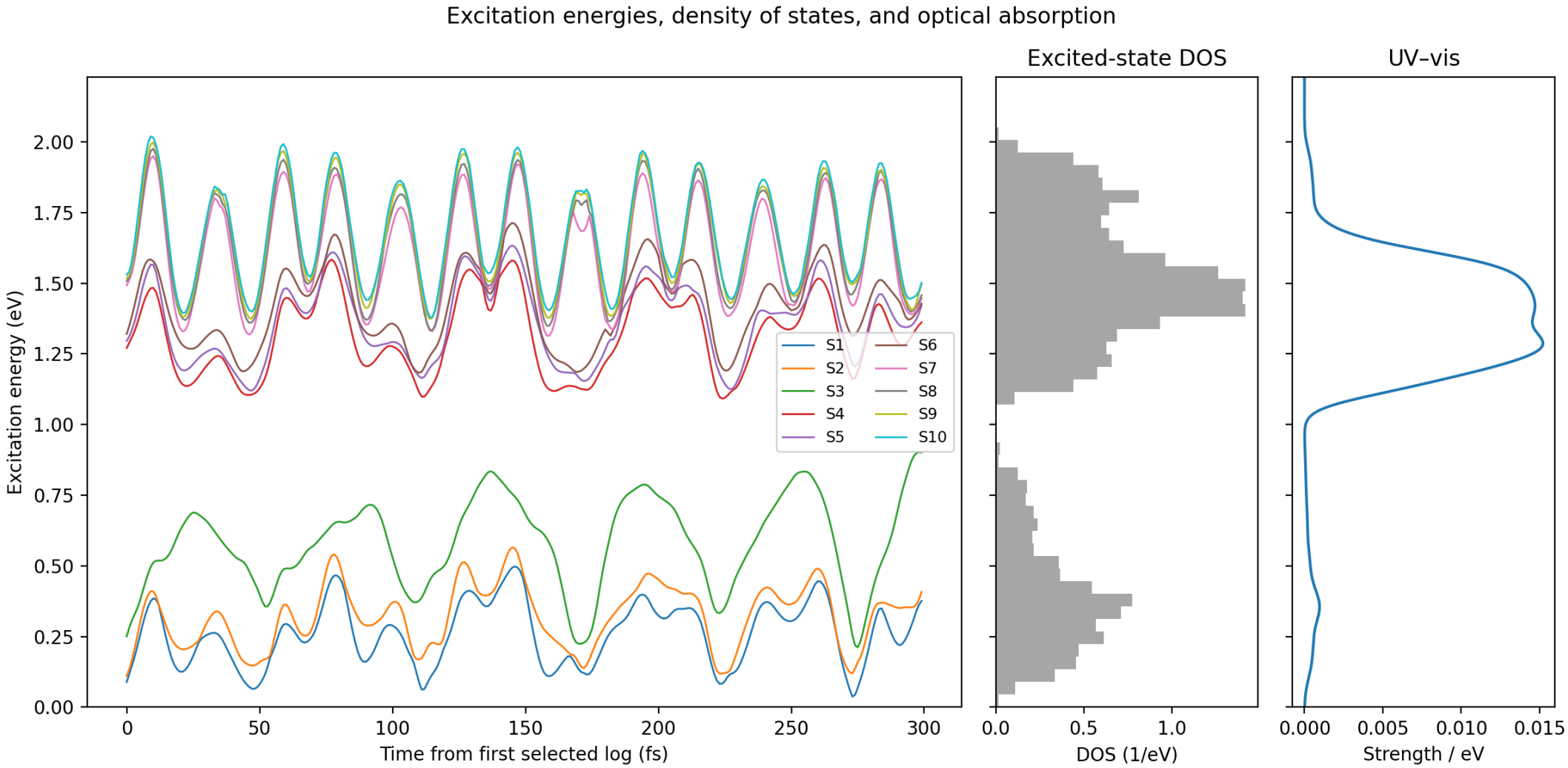
$$A(E) = \frac{1}{N_t}\sum_{n=0}^{N_t-1}\sum_{a\in\mathcal{A}} f_{na} G_\sigma(E-\epsilon_{na}).$$



$$\rho_b^{\mathrm{DOS}} = \frac{\sum_{n,a\in\mathcal{A}} \mathbf{1}[\epsilon_{na}\in B_b]}{N_t,\ |\mathcal{A}|,\Delta E_b}.$$

Step 3 plotting: Energy vs time, DOS, and UV-VIS (current plots; PBE)

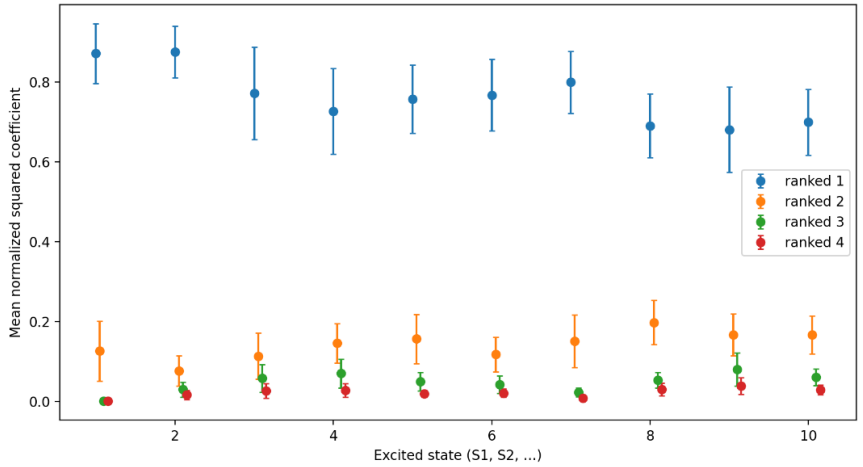
University at Buffalo
The State University of New York

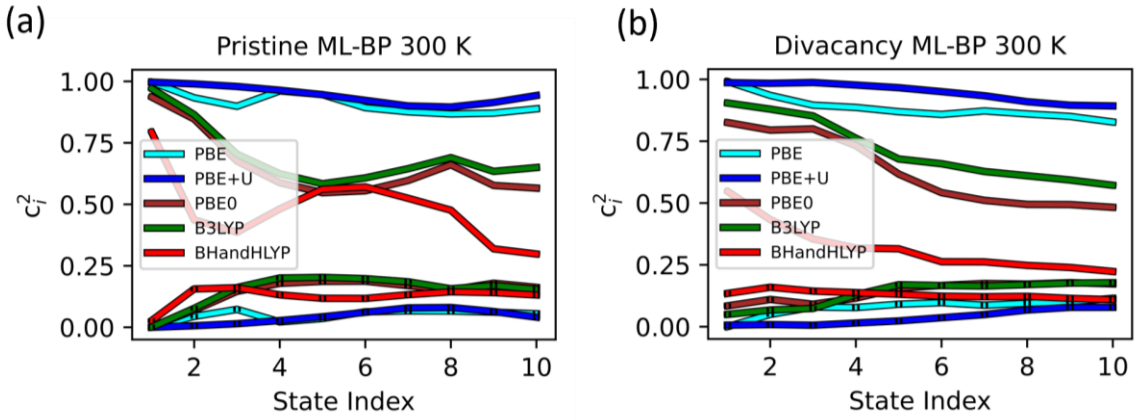


Step 3 plotting: Understanding composition of excited states

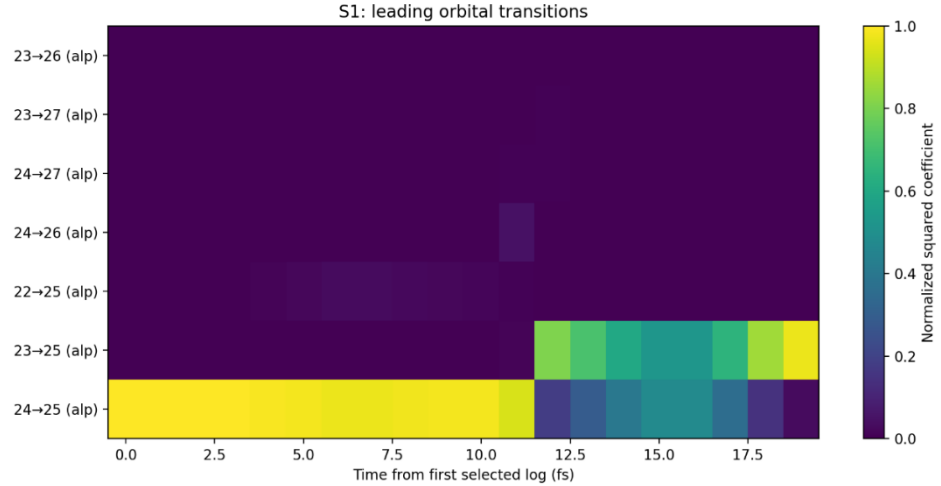
University at Buffalo
The State University of New York

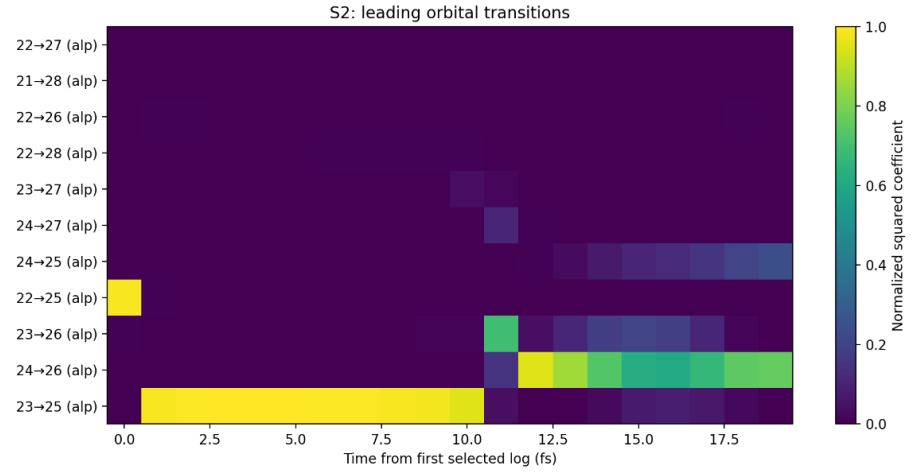
Average fraction of leading (or first printed) single-particle excitation in each excited state





Fraction of leading single-particle transitions in the time-evolving composition of each excited state

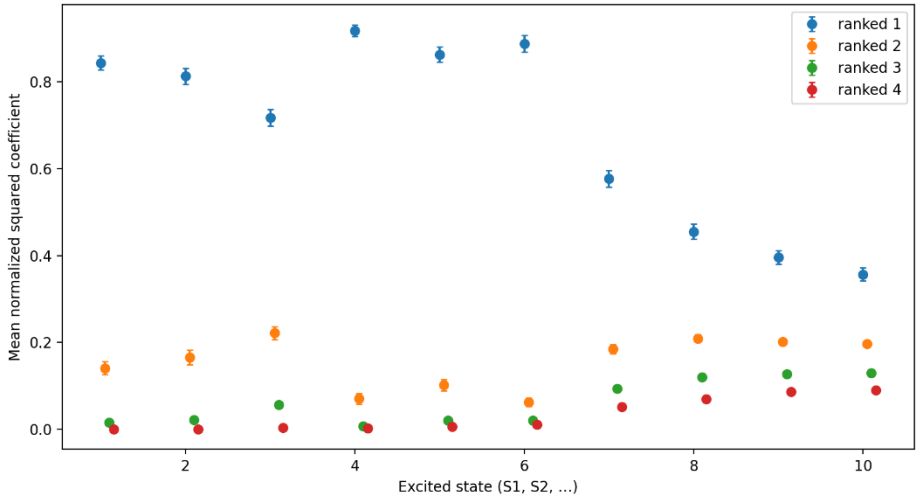


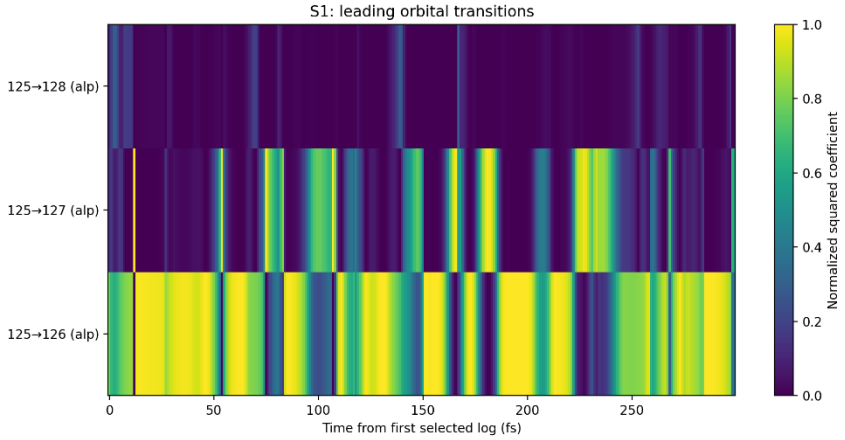


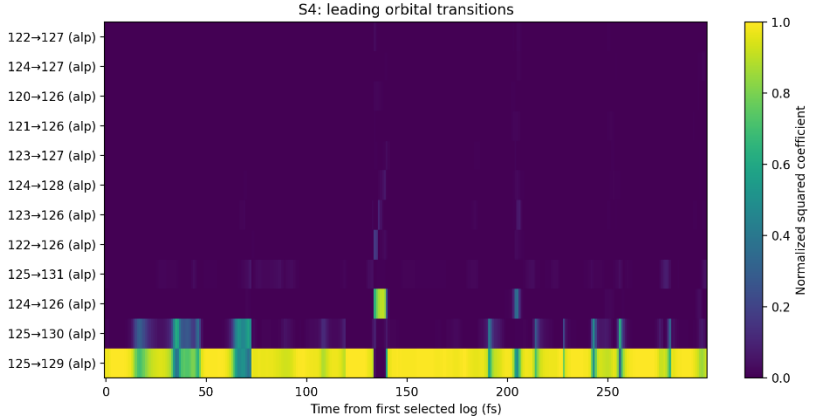
Zabihi, H.; Zhang, Q.; Khayati, G. R.; Irannejad, A.; Akimov, A. V. *JCTC* **2025** 21, 7991-8009

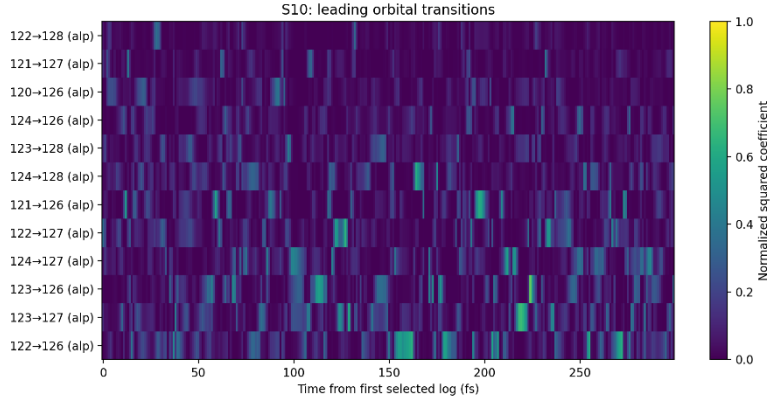
Step 3 plotting: Understanding composition of excited states (current plots; PBE)

University at Buffalo
The State University of New York



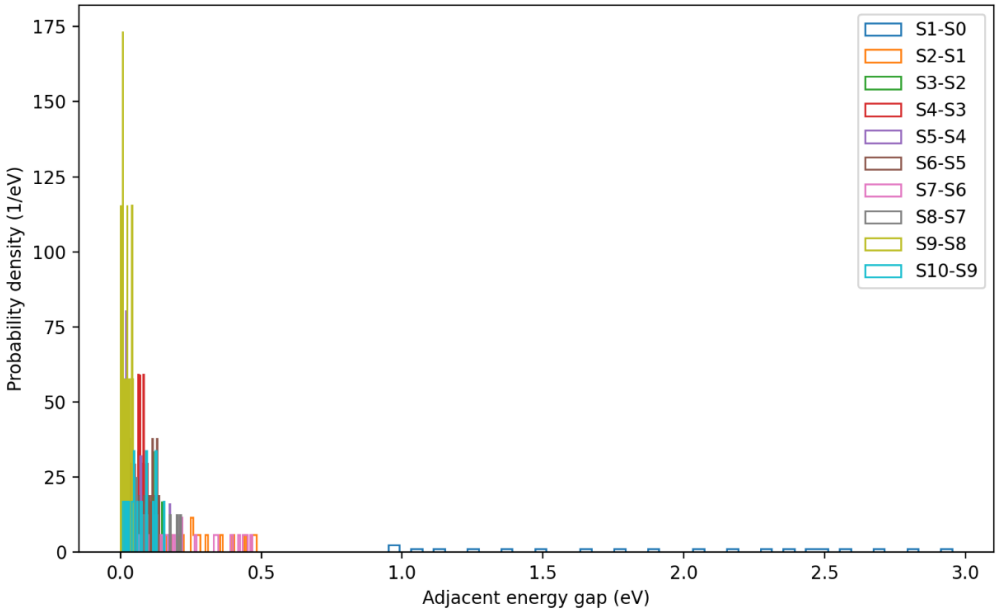






Step 3 plotting: Gap distributions

University at Buffalo
The State University of New York

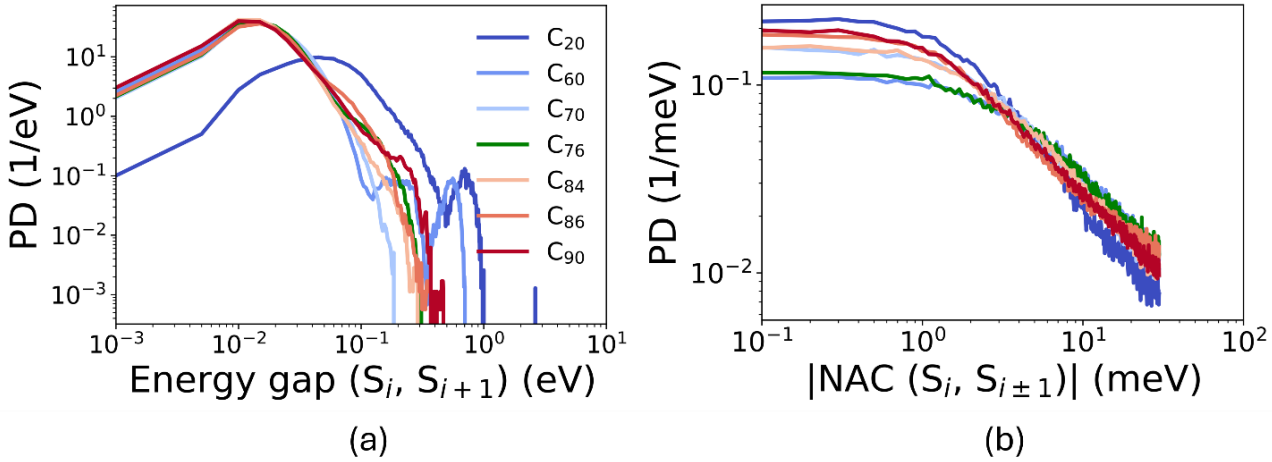


$$g_a(t_n) = E_{a+1}(t_n) - E_a(t_n)\,.$$

bin $B_b = [x_b, x_{b+1})$ of width Δx_b,

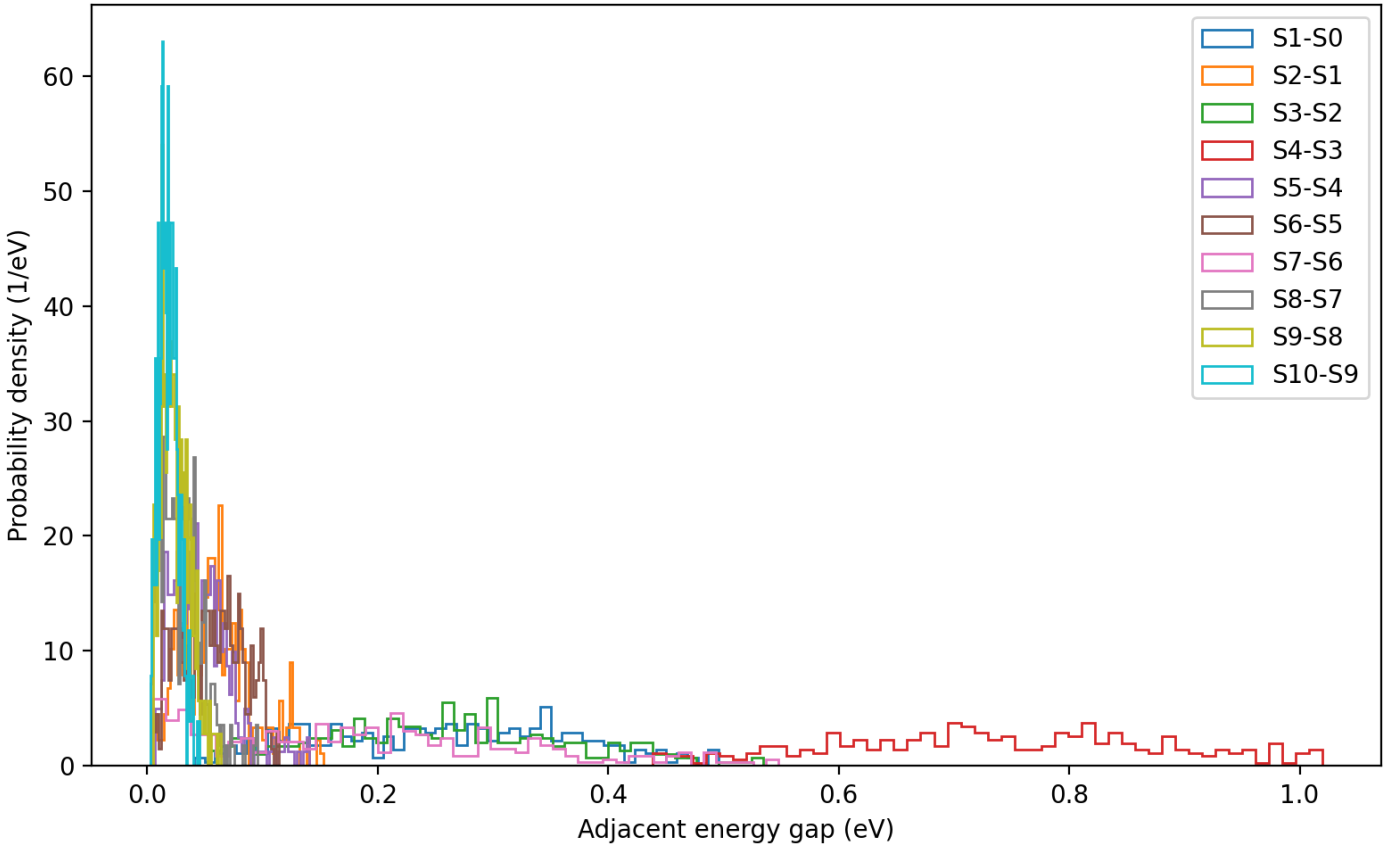
$$\rho_b = \frac{\sum_{n=0}^{N_t-1} \mathbf{1}[g_a(t_n) \in B_b]}{N_t, \Delta x_b}, \qquad \sum_b \rho_b \Delta x_b = 1.$$

Yasin, K.; Shakiba, M.; AVA *JPCL* **2026** 17, 2812



Step 3 plotting: Gap distributions (current scripts; PBE)

University at Buffalo
The State University of New York

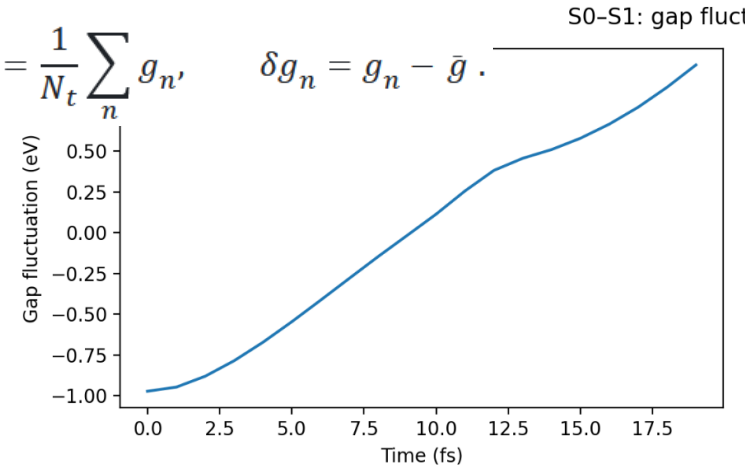


Step 3 plotting: Gap fluctuation, ACF, Influence spectrum and Spectral Density

University at Buffalo
The State University of New York

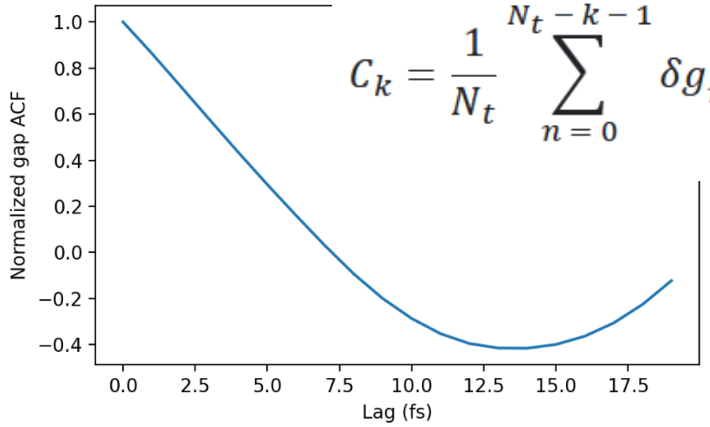
Gap fluctuation

$$g_n = E_j(t_n) - E_i(t_n), \qquad \bar{g} = \frac{1}{N_t}\sum_n g_n, \qquad \delta g_n = g_n - \bar{g}\ .$$



Normalized gap autocorrelation function

$$C_k = \frac{1}{N_t}\sum_{n=0}^{N_t-k-1} \delta g_n \delta g_{n+k}, \qquad \hat{C}_k = \frac{C_k}{C_0}.$$

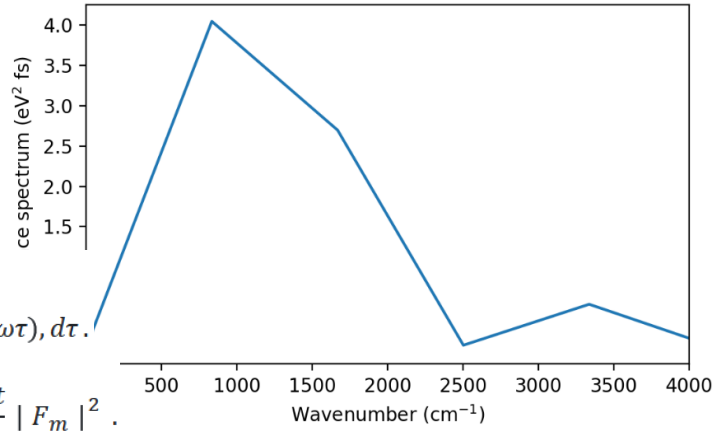


Influence spectrum

How much energy-gap fluctuation occurs at each frequency, at the sampled temperature?

$$I(\omega) = \int^{\infty} C(\tau)e^{-i\omega\tau}, d\tau = 2\int^{\infty} C(\tau)\cos(\omega\tau), d\tau\ .$$

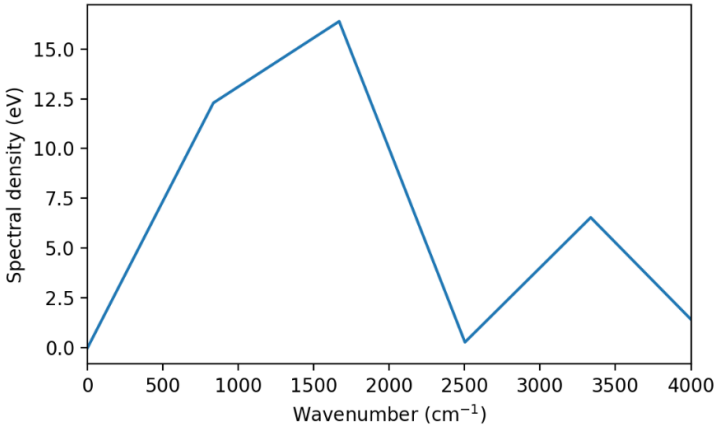
$$I_m = \Delta t\left[C_0 + 2\sum_{k=1}^{N_t-1} C_k \cos(\omega_m k\,\Delta t)\right] = \frac{\Delta t}{N_t}\,|\,F_m\,|^2\ .$$



Spectral density

How strongly does the electronic energy gap couple to bath modes at each frequency?

$$J_m = \frac{\beta_T \omega_m}{2} I_m, \qquad \beta_T = \frac{1}{k_B T}.$$

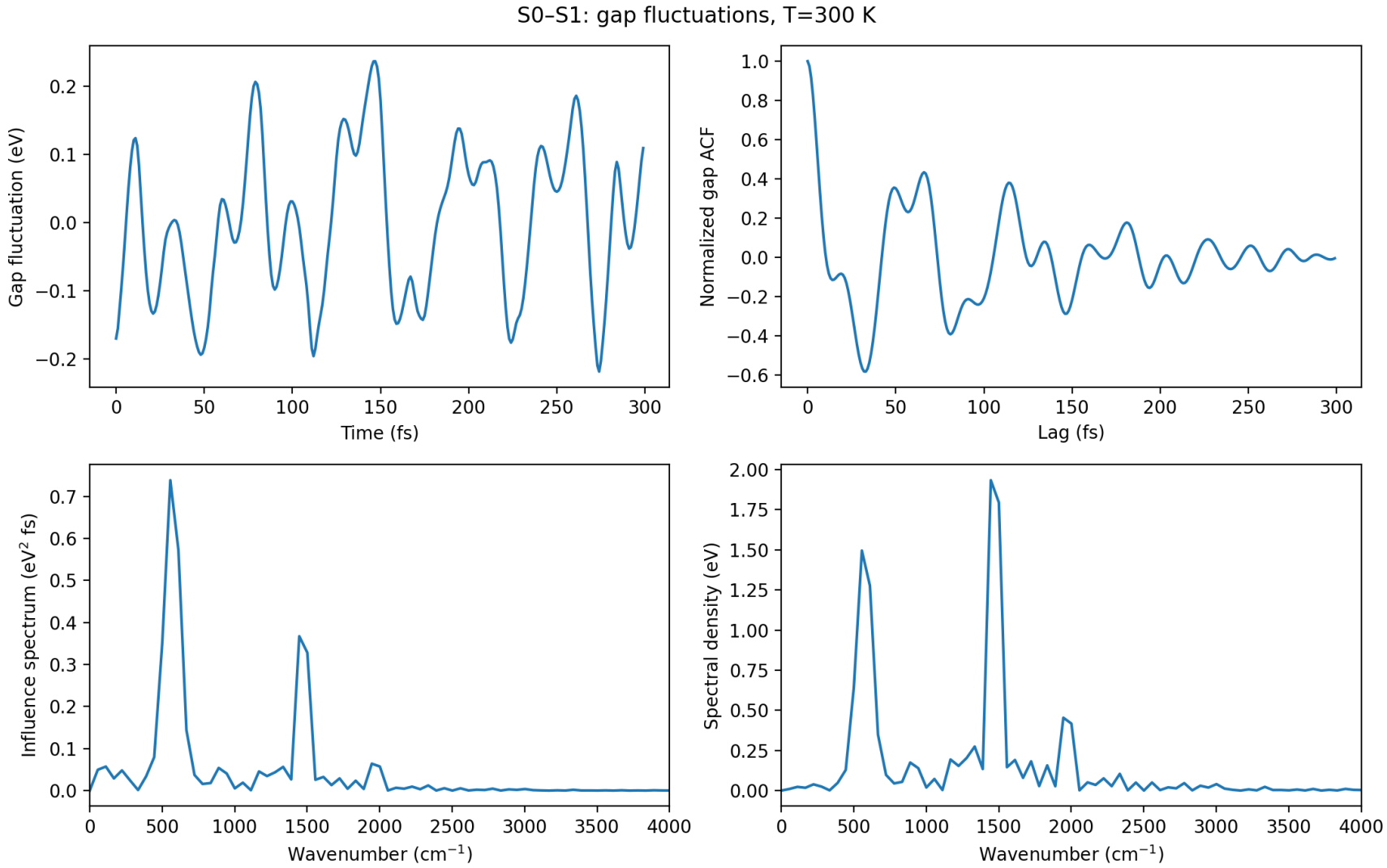


$$F_m = \sum_{n=0}^{N_t-1} \delta g_n e^{-2\pi i m n / L}, \qquad \omega_m = \frac{2\pi m}{L\,\Delta t}, \qquad I_m = \frac{\Delta t}{N_t}\,|\,F_m\,|^2\ ,$$

Stronger thermal motion can increase $I(\omega)$ even when the underlying coupling is unchanged

Step 3 plotting: Gap fluctuation, ACF, Influence spectrum and Spectral Density (current plots; PBE)

University at Buffalo
The State University of New York



Step 3 plotting: NAC distributions

University at Buffalo
The State University of New York

Scalar nonadiabatic couplings $$d_{ij} = \langle \Psi_i | \frac{\partial}{\partial t} \Psi_j \rangle \approx \frac{\langle \Psi_i(t - \Delta t) | \Psi_j(t) \rangle - \langle \Psi_i(t) | \Psi_j(t - \Delta t) \rangle}{2\Delta t}$$

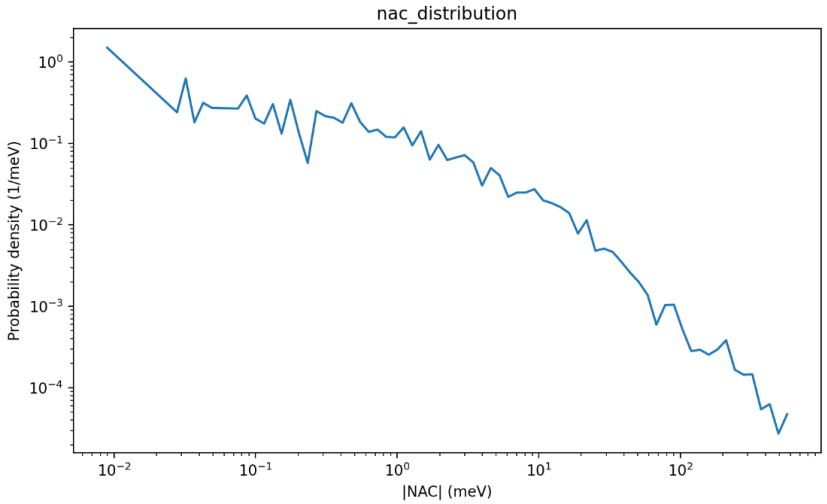
NAC distributions

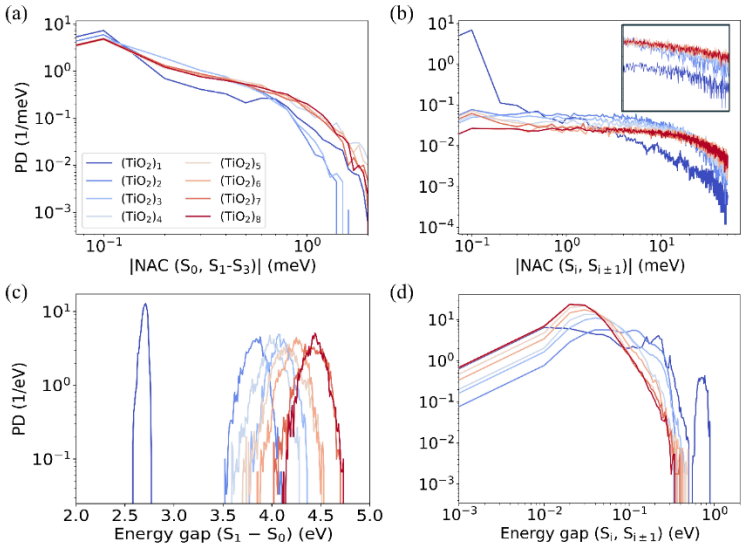
$$x_{n,ab} = |\hbar d_{ab}(t_n)|$$

$$\rho_j^{\mathrm{NAC}} = \frac{\left|\{(n,a,b): (a,b) \in \mathcal{P}, x_{n,ab} > 0, x_{n,ab} \in B_j\}\right|}{N_t \Delta x_j}.$$

NAC_PAIRS="all“: $\mathcal{P} = (a,b): a < b$

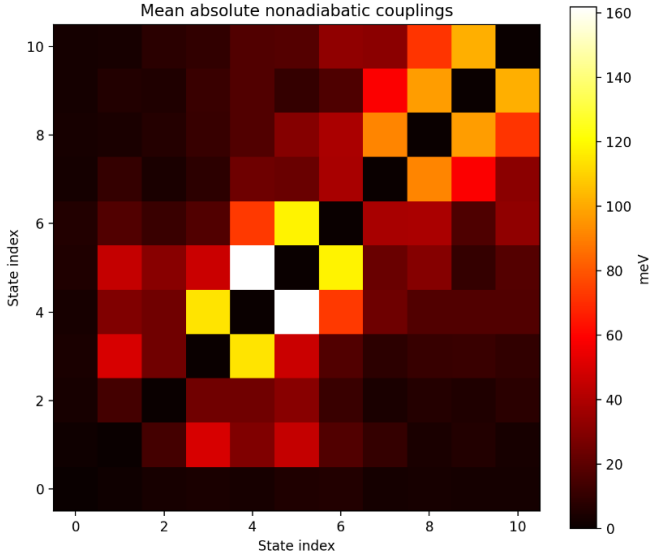
NAC_PAIRS="adjacent“: $\mathcal{P} = (a, a+1)$





NAC heatmaps

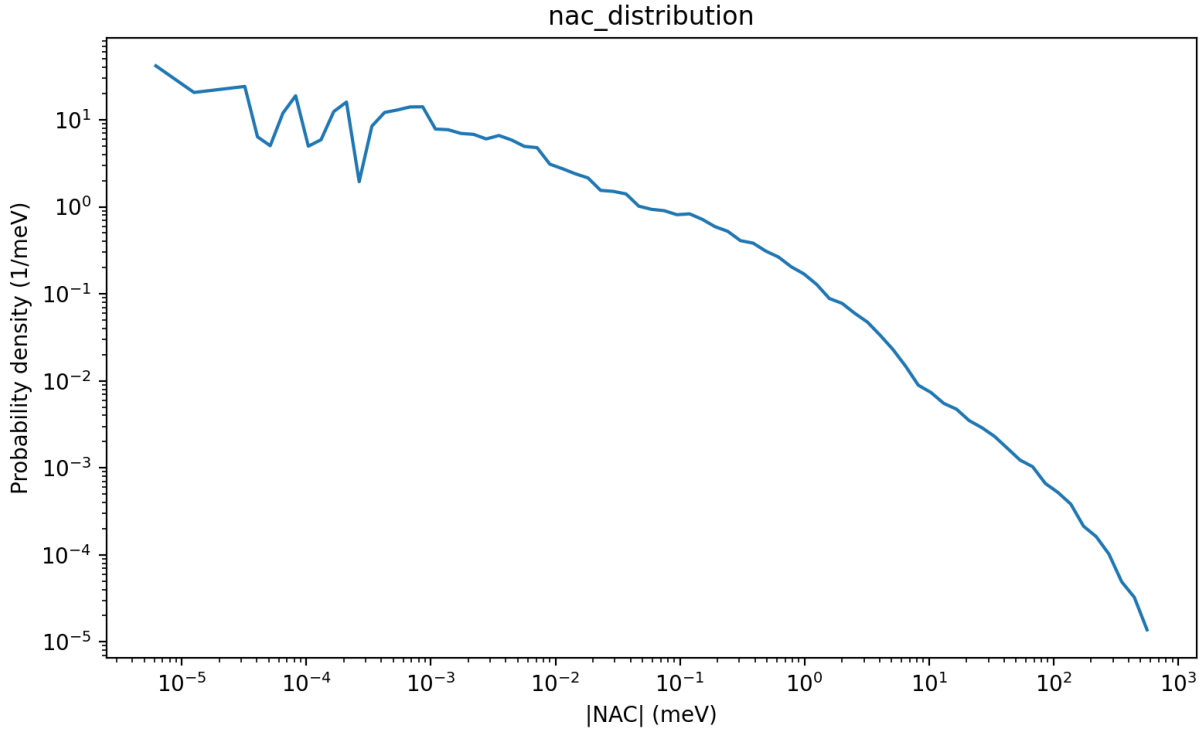
$$A_{ab} = \frac{1}{K} \sum_{n \in \mathcal{I}} |\hbar d_{ab}(t_n)|, \qquad A_{aa} = 0,$$

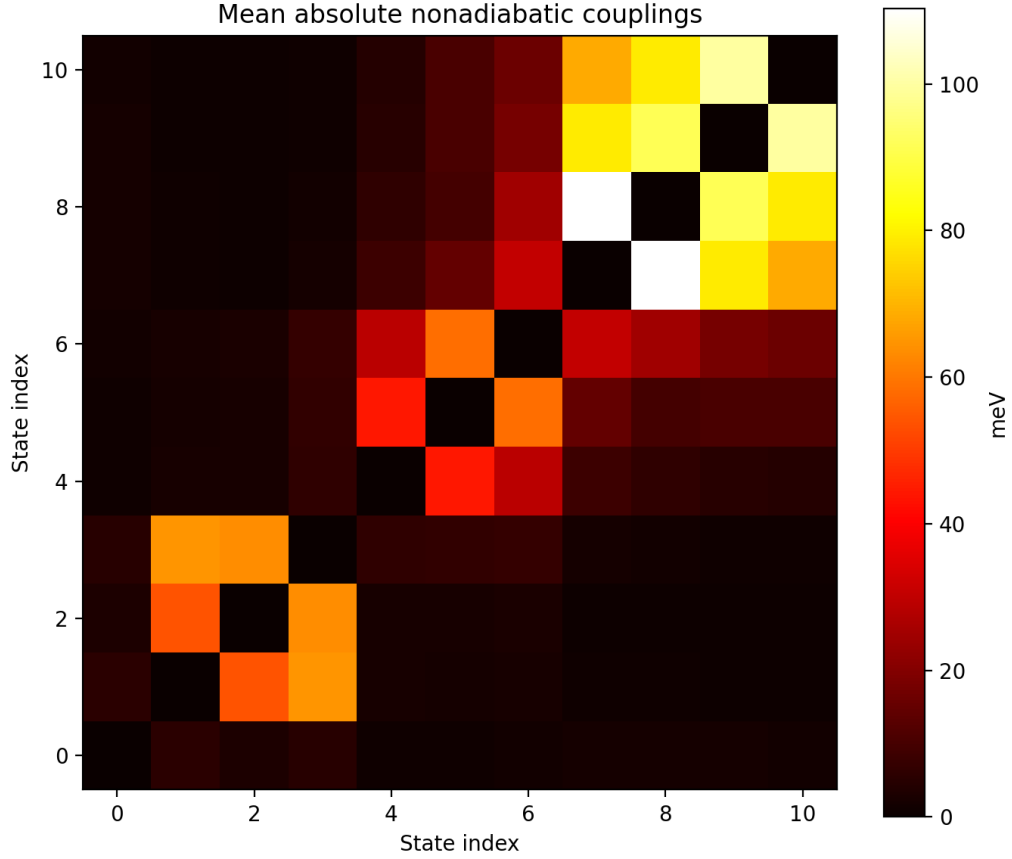


Recio-Poo, M.; Rotteger, C. H.; Illas, F.; Bromley, S. T.; Morales-Garcia, A.; Sayres, S.; Akimov, A. V.; *J. Am. Chem. Soc.* **2025**, 147, 44, 40900–40909

Step 3 plotting: NAC distributions (current plots; PBE)

University at Buffalo
The State University of New York





Step 4: TD-SE integration

University at Buffalo
The State University of New York

$\Psi(t) = |\boldsymbol{\psi}\rangle c_A$ - overall time-dependent wavefunction

Dynamical basis (adiabatic states):

$$|\boldsymbol{\psi}\rangle = (\psi_1, \psi_2, \ldots \psi_{N_A})$$

$$\boldsymbol{c}_A = (c_1, c_2, \ldots, c_{N_A})^T$$

Electronic Dynamics – TDSE:

$$i\hbar \dot{c}_i = \sum_j [E_i \delta_{ij} - i\hbar d_{ij}] c_j$$

vibronic Hamiltonian: $H_{ij}^{vib} = E_i \delta_{ij} - i\hbar d_{ij}$

Formal solution (NAC-based) :

$$\boldsymbol{c}_A(t_{n+1}) = e^{-i\boldsymbol{H}^{vib}\Delta t} \boldsymbol{c}_A(t_n)$$

- needs phase tracking, state tracking
- sensitive/inaccurate when NACs are spiky

Time-overlaps: $\boldsymbol{S}_n = \langle \boldsymbol{\psi}(\boldsymbol{R}(t_{n-1})) | \boldsymbol{\psi}(\boldsymbol{R}(t_n)) \rangle$

Singular value decomposition (SVD):

$$\boldsymbol{S}_{A,n} = \boldsymbol{U}_{A,n} \boldsymbol{\Sigma}_{A,n} \boldsymbol{V}_{A,n}^{+}$$

singular values: $\boldsymbol{\Sigma}_{A,n} = diag(\sigma_1, \sigma_2, \ldots, \sigma_{N_A})$

$\sigma_i \to 0$ means losing rank (the corresponding eigenvector is linearly-dependent on all other eigenvectors, matrix is ill-conditioned, redundant directions)

Local diabatization (LD):

$$\boldsymbol{T}_{A,n} = \boldsymbol{U}_{A,n} \boldsymbol{V}_{A,n}^{+}$$

Basically: $\boldsymbol{\Sigma}_{A,n} \to \boldsymbol{I}_A$ - retaining all directions

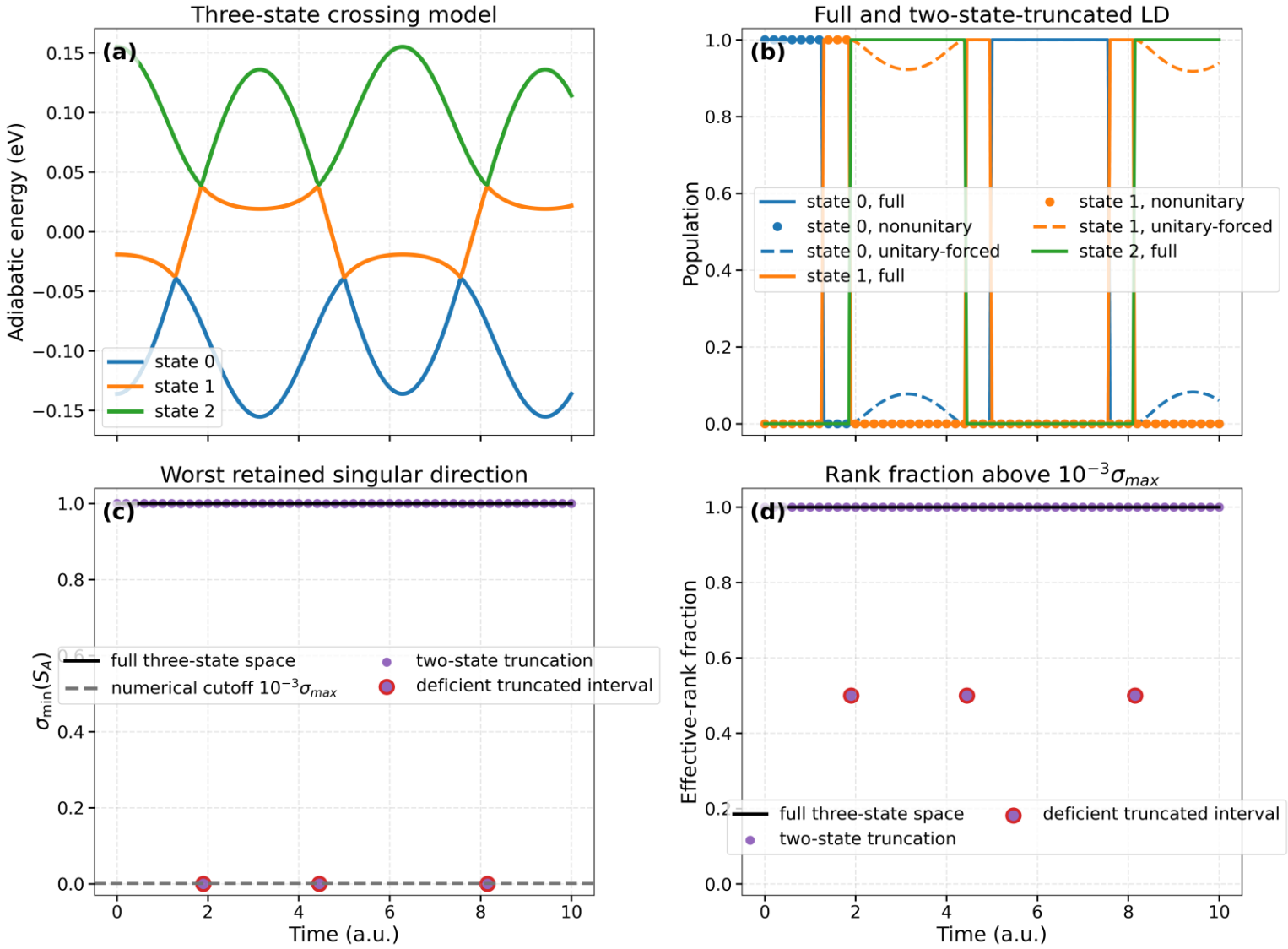
$$\boldsymbol{c}_A(t_{n+1}) = e^{-iE_A(t_{n+1})\Delta t/2} \boldsymbol{T}_{A,n}^{+} e^{-i\boldsymbol{E}_A(t_n)\Delta t/2} \boldsymbol{c}_A(t_n)$$

- Takes care of state reordering and phase mismatch
- Very robust and allows larger Δt

Where LD has a hard time: a 3-state toy model

Akimov A.V. *JPCL* **2026** https://doi.org/10.1021/acs.jpclett.6c03072

University at Buffalo
The State University of New York



3-state model (complete) is fine!

Truncating it to 2 states is bad – because singular values at sharp green/orange intersections are zero – not full rank

The consequence: artificial population dynamics in the truncated space

There is no cure: but we can detect this case

And not expect the dynamics in the truncated space be correct.

LD needs a suitable dynamical basis: SVD-based metrics

Akimov A.V. *JPCL* **2026** https://doi.org/10.1021/acs.jpclett.6c03072

University at Buffalo
The State University of New York

Thresholds for the retained singular values

$$\tau_{A,n} = \max\left(\tau_{\text{abs}}, \tau_{\text{rel}}\sigma_{A,n,1}\right)$$

$$\tau_{\text{abs}} = 10^{-12} \qquad \tau_{\text{rel}} = 10^{-3}$$

Trajectory-wide numerical support margin

$$m_A = \min_n \frac{\sigma_{A,n,N_A}}{\tau_{A,n}}$$

A candidate dynamical basis A is considered as numerically supported only when $m_A > 1$

Nominal rank N_A

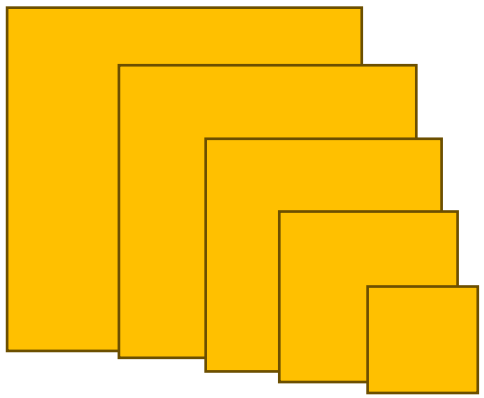
True rank $r_{A,n} = \#\{k: \sigma_{A,n,k} > \tau_{A,n}\}$

Effective rank $f_{A,n} = r_{A,n}/N_A$.

Trajectory-wide upper bound of the rank

$$r_* = \min_n r_n$$

Algorithm: systematically shrink candidate space A and check the rank/support margin



$N_A = 6 \Rightarrow r_A = 4 \Rightarrow$ **Bad, reject**

$N_A = 5 \Rightarrow r_A = 4 \Rightarrow$ **Bad, reject**

$N_A = 4 \Rightarrow r_A = 4 \Rightarrow$ **Good, accept**

$N_A = 3 \Rightarrow r_A = 2 \Rightarrow$ **Bad, reject**

$N_A = 2 \Rightarrow r_A = 2 \Rightarrow$ **Good, accept**

Step 4: Trajectory Surface Hopping (TSH)

University at Buffalo
The State University of New York

>> vi recipes/fssh_nbra.py

```
#============ Surface hop proposal methodology =================
#dyn_general.update({"tsh_method":-1 }) # adiabatic dynamics, no hops [ default ]
dyn_general.update({"tsh_method":0 }) # Fewest Switches Surface Hopping (FSSH)
#dyn_general.update({"tsh_method":1 }) # Global Flux Surface Hopping (GFSH)
#dyn_general.update({"tsh_method":2 }) # Markov-State Surface Hopping (MSSH)
#dyn_general.update({"tsh_method":3, "rep_lz":0 }) # Landau-Zener (LZ) options
#dyn_general.update({"tsh_method":4, "rep_lz":0 }) # Zhu-Nakamura (ZN) options
#dyn_general.update({"tsh_method":5 }) # DISH
#dyn_general.update({"tsh_method":6 }) # MASH
#dyn_general.update({"tsh_method":7 }) # FSSH2
#dyn_general.update({"tsh_method":8 }) # FSSH3
#dyn_general.update({"tsh_method":9 }) # GFSH (original)
```

Hop proposal:

Tully, J. C. *J. Chem. Phys.* **1990**, *93*, 1061

$$\sigma(x) = xH(x) = \begin{cases} x, x \geq 0 \\ 0, otherwise \end{cases}$$

$$P_{i\to j}^{FSSH} = \sigma\left(\frac{2Re\left(\rho_{ij}^{*} d_{ij}\right)\Delta t}{\rho_{ii}}\right)$$

$$\rho_{ij} = c_i c_j^{*}$$

Hop acceptance:

Prezhdo, O. V.; Duncan, W. R.; Prezhdo, V. V. *Prog. Surf. Sci.* **2009**, *84*, 30

$$P_{i\to j}^{acc} = \min\left(1, \exp\left(-\frac{E_j - E_i}{k_B T}\right)\right)$$

- Instead of the energy conservation condition
- No velocity rescaling

```
#============ Types of surface hopping acceptance and momenta rescaling options ================
#dyn_general.update({"hop_acceptance_algo":10, "momenta_rescaling_algo":100 })  # accept and rescale based on total energy, do not reverse on frustrated
#dyn_general.update({"hop_acceptance_algo":10, "momenta_rescaling_algo":101 })  # accept and rescale based on total energy, reverse on frustrated
#dyn_general.update({"hop_acceptance_algo":20, "momenta_rescaling_algo":200 })  # accept and rescale based on NAC vectors, do not reverse on frustrated
#dyn_general.update({"hop_acceptance_algo":20, "momenta_rescaling_algo":201 })  # accept and rescale based on NAC vectors, reverse on frustrated
#dyn_general.update({"hop_acceptance_algo":21, "momenta_rescaling_algo":200 })  # accept and rescale based on force differences, do not reverse on frustrated
#dyn_general.update({"hop_acceptance_algo":21, "momenta_rescaling_algo":201 })  # accept and rescale based on force differences, reverse on frustrated
dyn_general.update({"hop_acceptance_algo":31, "momenta_rescaling_algo":0 })  # quantum Boltzmann acceptance; no rescaling
#dyn_general.update({"hop_acceptance_algo":32, "momenta_rescaling_algo":0 })  # classical Maxwell-Boltzmann acceptance; no rescaling
```

Step 4: Accounting for Electronic Decoherence

University at Buffalo
The State University of New York

SDM Granucci, G.; Persico, M. *J. Chem. Phys.* **2007**, *126*, 134114.

gradually change the amplitudes $C_i' = C_i \exp\left(-\frac{\Delta t}{\tau_{if}}\right), \forall i \neq f$ $\quad C_f' = C_f \sqrt{\frac{1-\sum_{i\neq f}|C_i'|^2}{|C_f|^2}}$

Implicit: energy-based decoherence (**EDC**) is used: $\tau_{ij}^{EDC} = \frac{\hbar}{|E_i - E_j|}\left(\mathrm{C} + \frac{\epsilon}{E_{kin}}\right)$

mSDM (modified SDM) Smith, B.; Akimov, A. V. *JCP* **2019**, 151, 124107

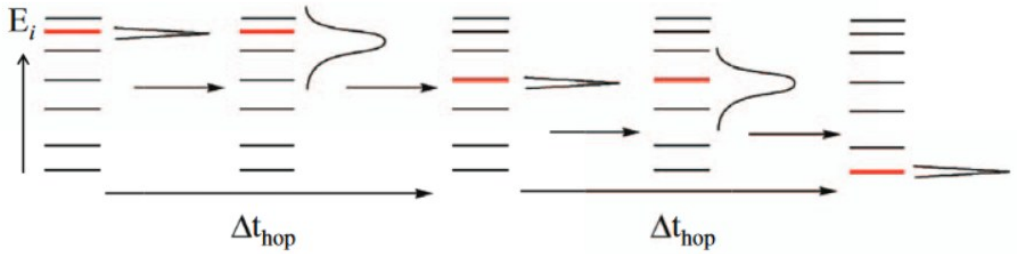
The only difference is that the decoherence times are defined as: $\tau_{ij}^{-1} = \sqrt{\frac{5\langle \delta E_{ij}^2 \rangle}{12\hbar^2}}$

Akimov, A. V.; Prezhdo, O. V. *JPCL,* **2013**, *4*, 3857

ID-A Nelson, T.; Fernandez-Alberti, S.; Roitberg, A. E.; Tretiak, S. *J. Chem. Phys.* **2013**, *138*, 224111.

Wavefunction reduction at every attempted hop

$C_f = 1, C_i = 0, \forall i \neq f$



Original DISH: Jaeger, H. M.; Fischer, S.; Prezhdo, O. V. *J. Chem. Phys.* **2012**, 137, 22A545

DISH_rev2023 (referred to as "DISH" later): Zhang, Q.; Shao, X.; Li, W.; Mi, W.; Pavanello, M.; Akimov, A. V. *J. Phys.: Condens. Matter* **2024**, *36*, 385901

Coherence interval for state i: $\frac{1}{\tau_i} = \sum_{j\neq i} \rho_{jj}\, r_{ij}$; $\rho_{jj} = |c_j|^2$; r_{ij} - pure dephasing time for the pair of states *i* and *j*

Conduct stochastic wavefunction collapses

State i decoheres: $\tilde{\Psi} \rightarrow \begin{cases} \Psi_i, & \text{with probability } |c_i|^2 \\ \sum_{j\neq i} \frac{c_j}{\sqrt{1-|c_i|^2}} \Psi_j, & \text{with probability } 1-|c_i|^2 \end{cases}$

Setting Decoherence Correction Methods

University at Buffalo
The State University of New York

>> vi recipes/fssh_nbra.py

Decoherence correction method

```
#========== Decoherence options =================
dyn_general.update({ "decoherence_algo":-1}) # no (additional) decoherence
#dyn_general.update({ "decoherence_algo":0}) # msdm
#dyn_general.update({ "decoherence_algo":1}) # IDA
#dyn_general.update({ "decoherence_algo":2}) # A-FSSH, not yet ready
#dyn_general.update({ "decoherence_algo":3}) # BCSH
#dyn_general.update({ "decoherence_algo":4}) # MF-SD
#dyn_general.update({ "decoherence_algo":5}) # SHXF
#dyn_general.update({ "decoherence_algo":6}) # MQCXF
#dyn_general.update({ "decoherence_algo":7}) # DISH, revised 2023
```

Decoherence time calculation method

```
#========== Decoherence times (MSDM, MF-SD, DISH variants) =================
# Manual times use the decoherence_rates matrix supplied by the simulation.
# For Schwartz options, first define A as the inverse-width MATRIX.
dyn_general.update({"decoherence_times_type":-1 }) # No decoherence times, infinite decoherence times
#dyn_general.update( { "decoherence_times_type":0 } )  # manual decoherence times
#dyn_general.update( { "decoherence_times_type":1, "decoherence_C_param": 1.0, "decoherence_eps_param":0.1 } )  # EDC + default params
#dyn_general.update( { "decoherence_times_type":2, "schwartz_decoherence_inv_alpha":A } ) # Schwartz version 1
#dyn_general.update( { "decoherence_times_type":3, "schwartz_decoherence_inv_alpha":A } ) # Schwartz version 2
```

Step 4: Hands On

University at Buffalo
The State University of New York

```
>> cd 4_step4
>> tar –xf data4.tar.bz2
```

First run: determining the suitable active space size

```
# -------------------- Example 4: Mg@C60, Mohammad's version (this tutorial) ----------
# New-style text files, archive data4.tar.bz2; input files are in data4/res.
# 4 states; dt = 1.0 fs; istep = 0, fstep = 300 (exclusive).
# If needed, extract in a terminal: tar -xf data4.tar.bz2
# This example is selected initially.
data_path = "data4/res"
istep = 0                          # First input file index
fstep = 300                         # Read through file 48
ISTATE = 1                         # Initially populated state, zero-based
DT = 1.0                           # Input frame spacing in fs
basis_size = None                  # First run: all available states
#basis_size = 4                     # Notebook example; select after the test
NSTEPS, NSTATES, St_adi, Ham_adi, Hvib_adi, NAC_adi, Basis_transform = read_files_new(
    data_path, istep, fstep, DT, basis_size)
```

```
# Keep this line active for the initial basis test.
# After choosing basis_size, comment it out to continue into sections 4-7.
sys.exit(0)
```

```
>> python NAMD_simulation.py --prefix MgC60_fssh
```

Example with current data

University at Buffalo
The State University of New York

Disregard these warnings

These sizes should be good

Danger zone

```
NAMD_plotting.py  README.md           data1         data2          data3          data4          reci
(libra) alexvakimov@Hegel:~/CCCT/Libra_CP2K_autumn_school_2026/4_step4$ python NAMD_simulation.py
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for std::vector<std::vector<ir
ocator<int> > > > already registered; second conversion method ignored.
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for boost::python::detail::cor
 std::allocator<std::vector<int, std::allocator<int> > > >, unsigned long, boost::python::detail::fir
ator<int> >, std::allocator<std::vector<int, std::allocator<int> > > >, false> > already registered;
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for std::vector<std::vector<fl
d::allocator<float> > > > already registered; second conversion method ignored.
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for boost::python::detail::cor
> >, std::allocator<std::vector<float, std::allocator<float> > > >, unsigned long, boost::python::det
std::allocator<float> >, std::allocator<std::vector<float, std::allocator<float> > > >, false> > alre
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for std::vector<std::vector<do
 std::allocator<double> > > > already registered; second conversion method ignored.
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for boost::python::detail::cor
le> >, std::allocator<std::vector<double, std::allocator<double> > > >, unsigned long, boost::python:
ble, std::allocator<double> >, std::allocator<std::vector<double, std::allocator<double> > > >, false
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for std::vector<std::vector<st
allocator<std::vector<std::complex<double>, std::allocator<std::complex<double> > > > > already regis
<frozen importlib._bootstrap>:241: RuntimeWarning: to-Python converter for boost::python::detail::cor
allocator<std::complex<double> > >, std::allocator<std::vector<std::complex<double>, std::allocator<s
:final_vector_derived_policies<std::vector<std::vector<std::complex<double>, std::allocator<std::comp
std::allocator<std::complex<double> > > > >, false> > already registered; second conversion method ig
Size   Accepted   Minimum rank   Minimum singular value
1 True 1 0.8735529
2 True 2 0.00018276226873906274
3 True 3 0.0173023343581283
4 True 4 0.8733857968996631
5 True 5 0.21594819387047123
6 True 6 0.020711882717378272
7 False 6 9.51341681822525e-05
8 False 7 9.944910275331594e-12
9 False 8 9.472220542673882e-12
10 False 9 4.540068196951977e-12
11 False 10 1.5363572878638929e-13
(libra) alexvakimov@Hegel:~/CCCT/Libra_CP2K_autumn_school_2026/4_step4$ ls
```

Step 4: Hands On

University at Buffalo
The State University of New York

Second run: select the suitable dynamical basis size and run the calculations

select the key simulation parameters: length, initial state, dt, etc.

```
# -------------------- Example 4: Mg@C60, Mohammad's version (this tutorial) ----------
# New-style text files, archive data4.tar.bz2; input files are in data4/res.
# 4 states; dt = 1.0 fs; istep = 0, fstep = 300 (exclusive).
# If needed, extract in a terminal: tar -xf data4.tar.bz2
# This example is selected initially.
data_path = "data4/res"
istep = 0                              # First input file index
fstep = 300                             # Read through file 48
ISTATE = 1                             # Initially populated state, zero-based
DT = 1.0                               # Input frame spacing in fs
#basis_size = None                      # First run: all available states
basis_size = 4                         # Notebook example; select after the test
NSTEPS, NSTATES, St_adi, Ham_adi, Hvib_adi, NAC_adi, Basis_transform = read_files_new(
    data_path, istep, fstep, DT, basis_size)
```

```
# Keep this line active for the initial basis test.
# After choosing basis_size, comment it out to continue into sections 4-7.
#sys.exit(0)
```

```
dyn_general = {"nsteps": NSTEPS, "ntraj": 250, "nstates": NSTATES,
               "dt": DT * units.fs2au, "nfiles": NSTEPS,
               "decoherence_rates": rates, "ave_gaps": gaps,
               "Temperature": 300.0, "progress_frequency": 0.1,
               "which_adi_states": list(range(NSTATES)),
               "which_dia_states": list(range(NSTATES)),
               "mem_output_level": 2,
               "properties_to_save": ["timestep", "time", "se_pop_adi", "sh_pop_adi"],
               "isNBRA": 0, "is_nbra": 0}
```

Step 4: Hands On

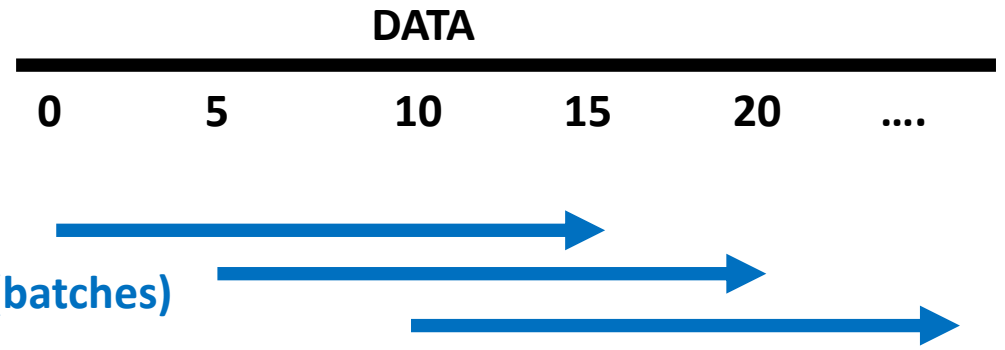
University at Buffalo
The State University of New York

Select the recipe defining the flavor of NA-MD simulation

```
# Select ONE recipe. Copy a file in recipes/ to make your own reusable recipe,
# edit its commented choices, then import it above and call its load() here.
# The notebook contrasts LD with NAC/Hvib propagation. Standard FSSH + LD uses
# tsh_method=0, decoherence_algo=-1, rep_tdse=1, electronic_integrator=2,
# state_tracking_algo=-1, and both separate phase corrections disabled.
# Its NAC-based example uses integrator=5, tracking=21, and ONE phase correction
# (do_phase_correction OR do_nac_phase_correction). Change these as a group in a
# copied recipe. FSSH2 + MSDM combines tsh_method=7, decoherence_algo=0 and
# decoherence_times_type=0 with the supplied rates; it is a separate recipe
# choice, not the default FSSH2 file's behavior.
fssh_nbra.load(dyn_general); prf = "FSSH"
#fssh_nbra_v2.load(dyn_general); prf = "FSSH"
#fssh_nbra_nac_v0.load(dyn_general); prf = "FSSH"
#fssh_nbra_nac_v1.load(dyn_general); prf = "FSSH"
#fssh_nbra_nac_v2.load(dyn_general); prf = "FSSH"
#fssh_nbra_nac_v3.load(dyn_general); prf = "FSSH"
#fssh_nbra_nac_v4.load(dyn_general); prf = "FSSH"
#fssh2_nbra.load(dyn_general); prf = "FSSH2"
#gfsh_nbra.load(dyn_general); prf = "GFSH"
#ida_nbra.load(dyn_general); prf = "IDA"
#msdm_nbra.load(dyn_general); prf = "MSDM"
#mash_nbra.load(dyn_general); prf = "MASH"
#dish_nbra.load(dyn_general); prf = "DISH"
#dish_rev2023_nbra.load(dyn_general); prf = "DISH_REV2023"
```

Starting points for the dynamics:

```
# Compare pooled/sequential results s
ICONDS = [0, 5, 10]
#ICONDS = list(range(0, 25, 5))
nthreads = 4
```



>> python NAMD_simulation.py --prefix MgC60_fssh

All the results will be written to the directory defined by the prefix

Step 4: Plotting Properties

University at Buffalo
The State University of New York

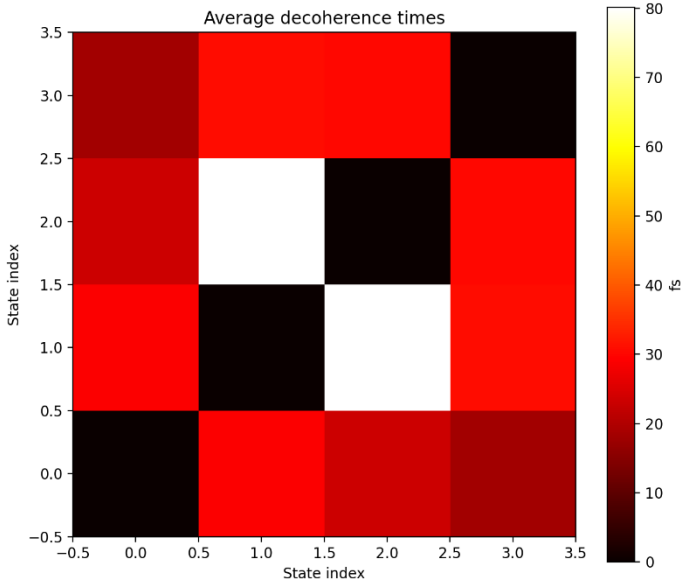
>> python NAMD_plotting.py --prefix MgC60_fssh

The input data for plotting is defined the prefix, the (additional) output will be written to the prefix directory too

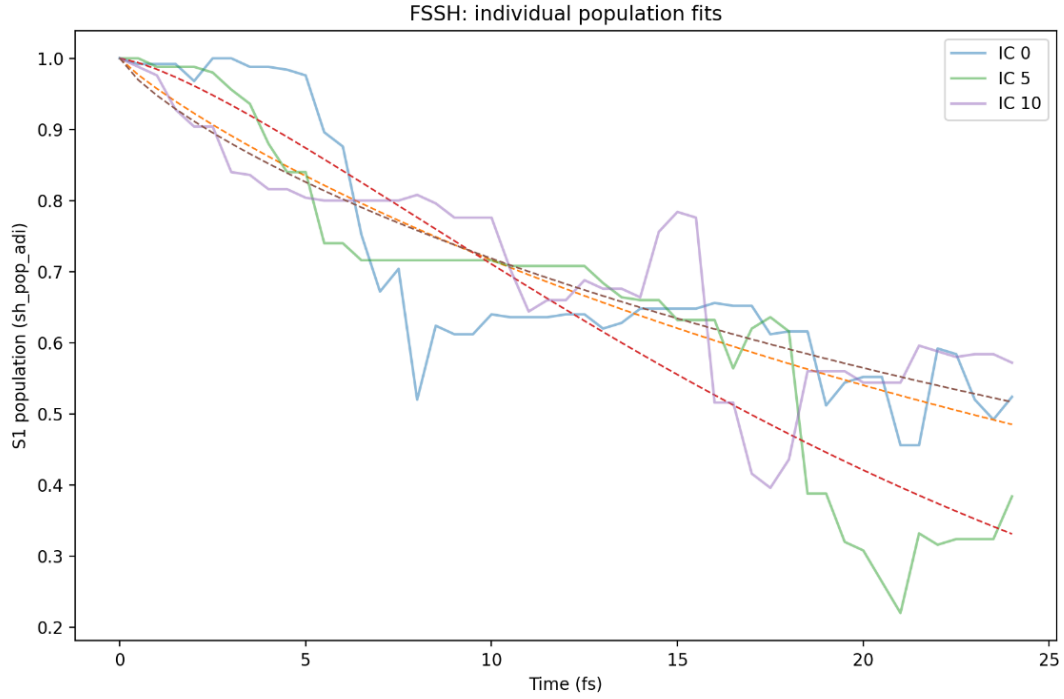
Pure dephasing times

$$\tau_{ij}^{-1} = \sqrt{\frac{5\langle \delta E_{ij}^2 \rangle}{12\hbar^2}}$$

Akimov, A. V.; Prezhdo, O. V. *JPCL,* **2013**, *4*, 3857



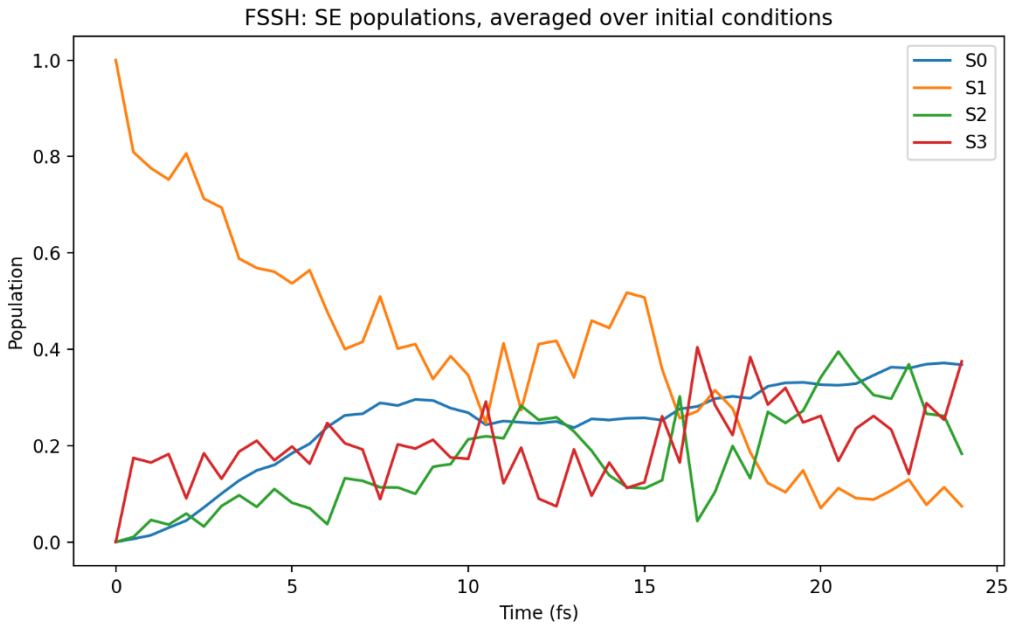
Individual populations and fits



Step 4: Populations

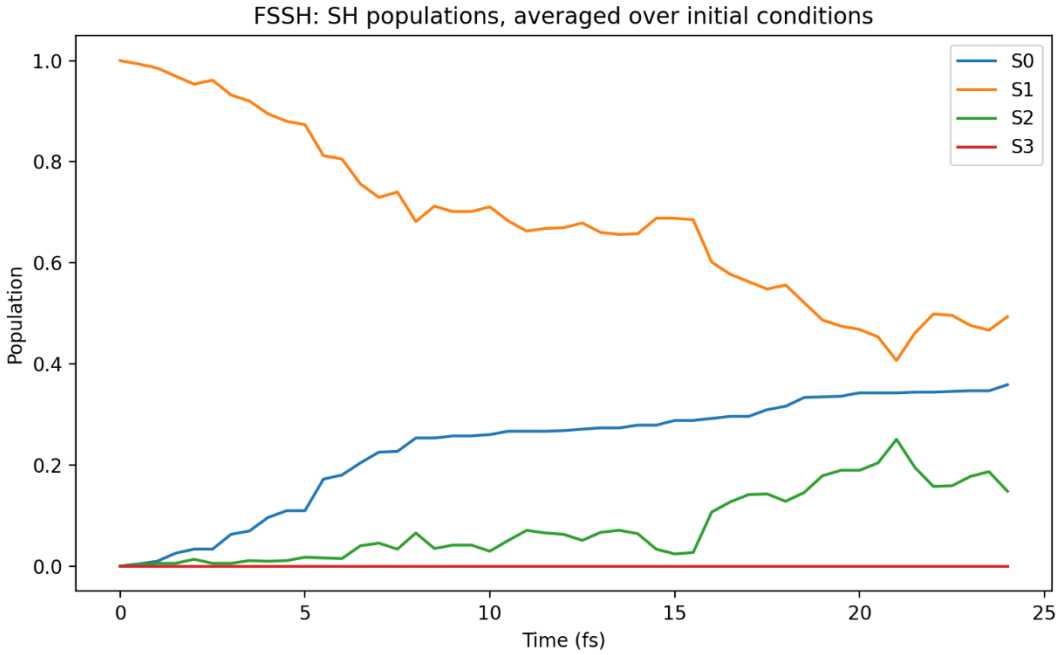
University at Buffalo
The State University of New York

SE populations



$$p_i^{SE}(t) = \frac{1}{N_{traj}} \sum_{k=1}^{N_{traj}} \left| c_i^{(k)}(t) \right|^2$$

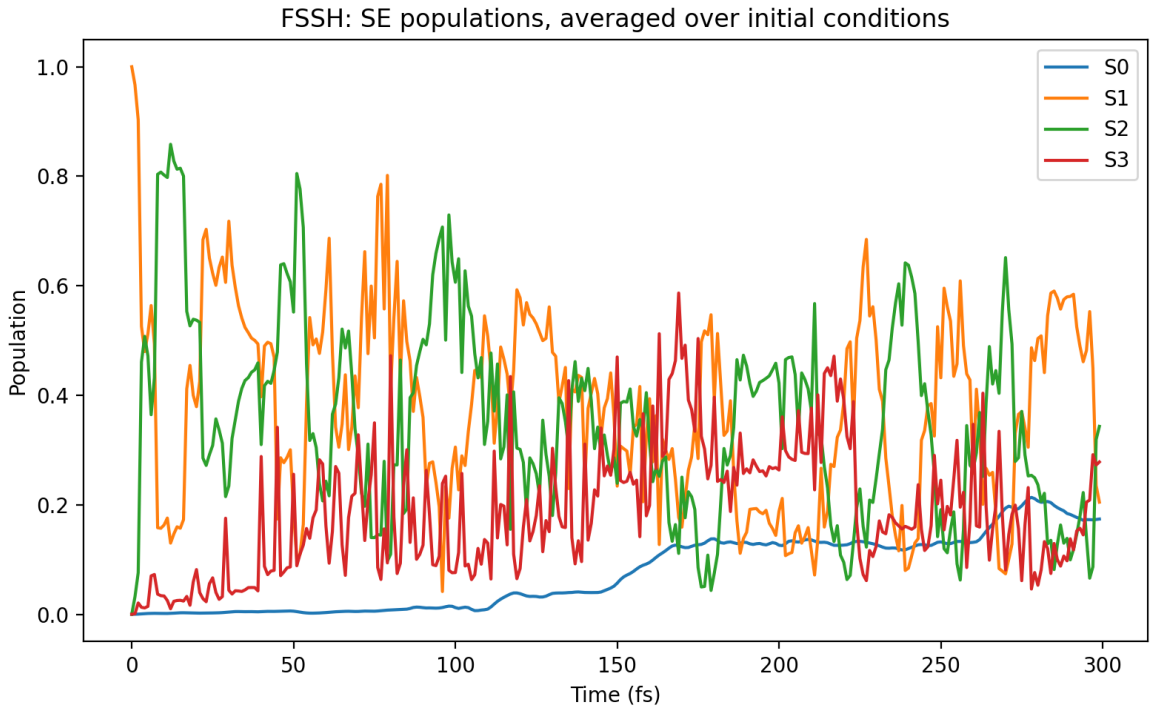
SH populations

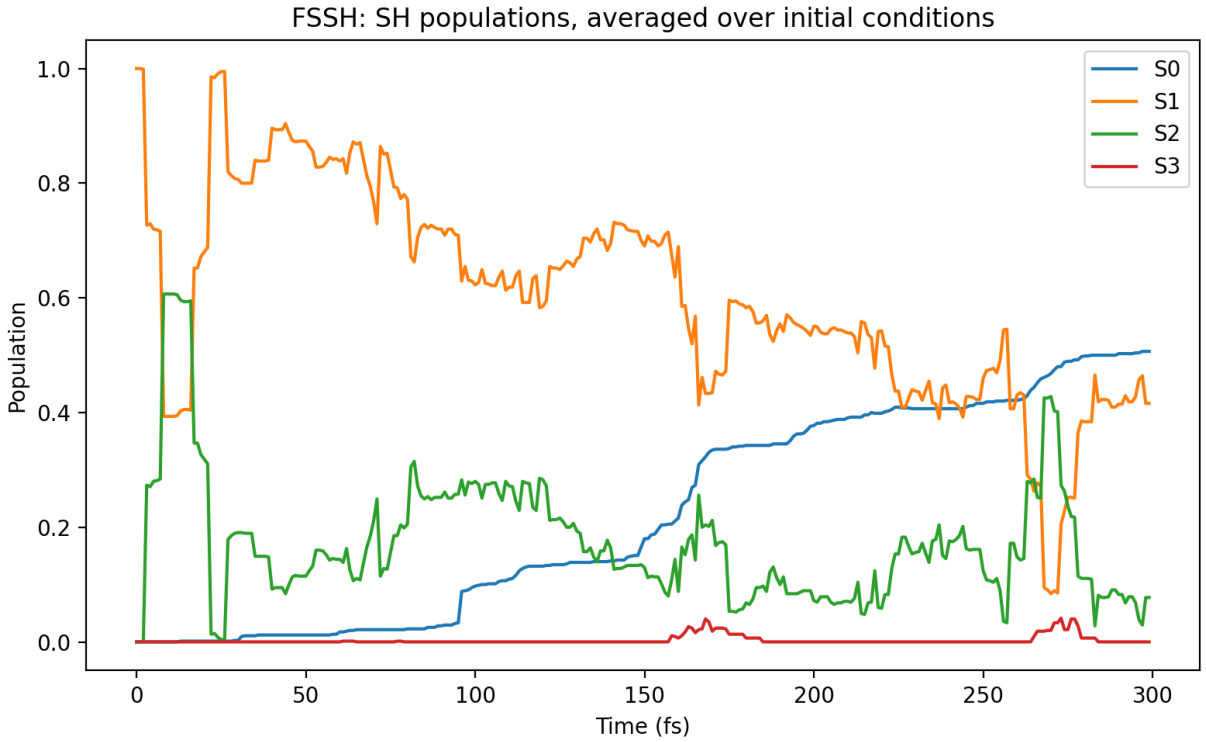


$$p_i^{SH}(t) = \frac{1}{N_{traj}} \sum_{k=1}^{N_{traj}} 1(a_k == i)$$

Step 4: Populations (data4)

University at Buffalo
The State University of New York





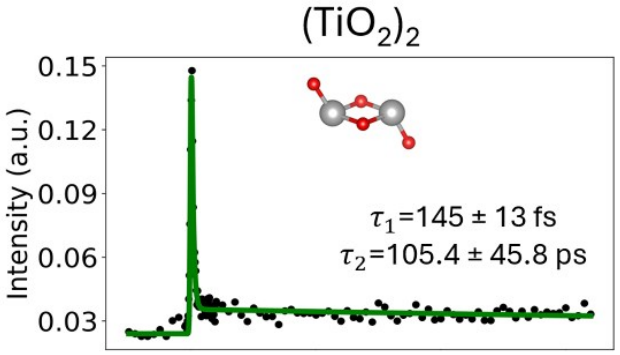
Step 4: TRPES and Shannon Entropy

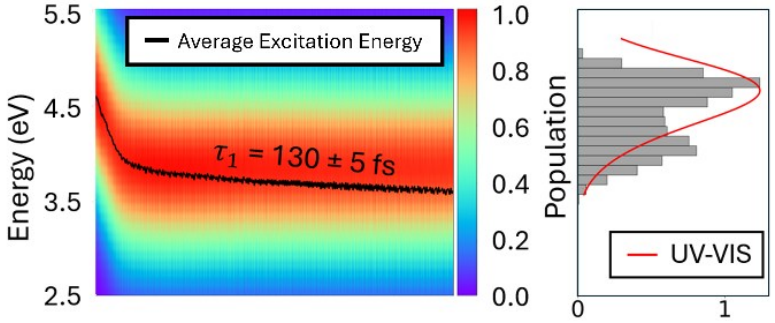
University at Buffalo
The State University of New York

TRPES: time-resolved population-energy spectrogram

Recio-Poo, M.; Rotteger, C. H.; Illas, F.; Bromley, S. T.; Morales-Garcia, A.; Sayres, S.; Akimov, A. V.; *J. Am. Chem. Soc.* **2025**, 147, 44, 40900–40909

$$I(t_i, E) = \frac{1}{N_{icond}\sigma\sqrt{2\pi}} \sum_{k=1}^{N_{icond}} P_{J,k}(t_i)\,\Delta E\,\exp\left(-\frac{\left(E - E_{J,k}(t_i)\right)^2}{2\sigma^2}\right)$$



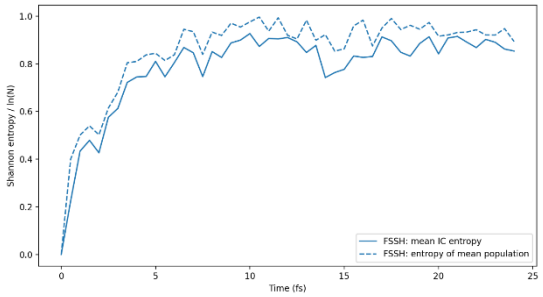


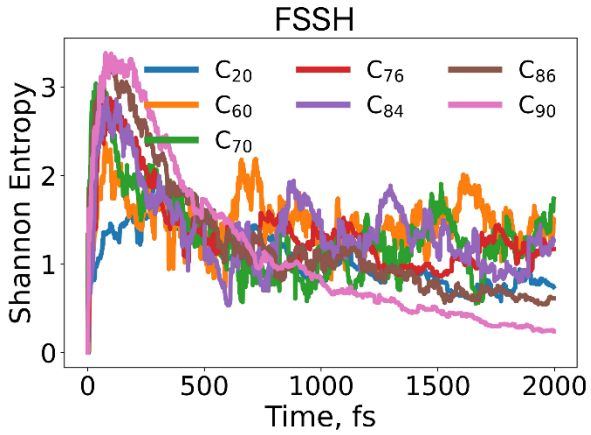
- experiment measures energy-related properties
- it is more invariant wrt state degeneracies/trivial crossings

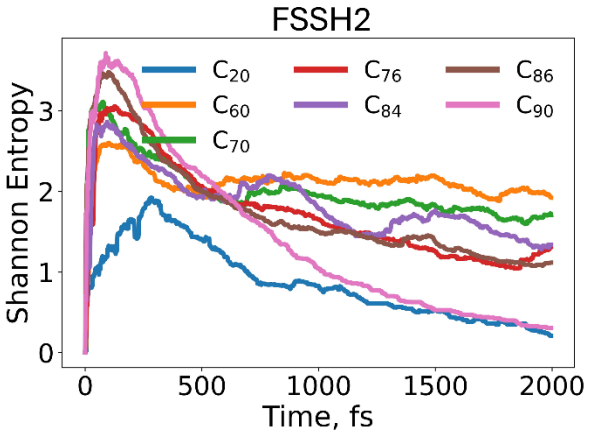
Shannon Entropy of quantum state

Yasin, K.; Shakiba, M.; AVA *JPCL* **2026** 17, 2812

$$S(t) = -\sum_{i=0}^{N_{st}-1} p_i(t)\ln(p_i(t))$$







- characterizes the rate of population “spreading”

Acknowledgements



Mohammad Shakiba (UB; Now: UC Merced)

Group members & collaborators:

- Kosar Yasin
- Layla Heidarizadeh
- Somesh Chandra
- Miguel Recio
- Daeho Han

Computing resources:

Center for Computational Research, University at Buffalo, The State University of New York

Funding:

NSF

Libra developers & Contributors



Organizers:

- Anna Henh
- Tejas Thorat
- CAU HPC team

Audience!